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A COMPOSITIONAL AND DIAGRAMMATIC SYNTAX FOR PARAMETRIC LINEAR PROGRAMMING

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Dissertação de Mestrado apresentada ao Programa de Pós-Graduação em Informática, Instituto de Computação e Instituto Tércio Pacitti de Aplicações e Pesquisas Computacionais, Universidade Federal do Rio de Janeiro, como requisito parcial à obtenção do título de Mestre em Informática.

Orientador: Prof. João Paixão, D.Sc.

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RESUMO

Esta dissertação investiga como problemas de otimização podem ser compreendidos dentro de um arcabouço composicional e diagramático unificado, fundamentado na teoria das categorias. A motivação central é a observação de que estruturas matemáticas essenciais em otimização — relações, funções convexas, formas quadráticas, poliedros e programas lineares — podem ser interpretadas como morfismos em categorias nas quais a composição corresponde à eliminação ou minimização de variáveis internas. A partir desta perspectiva, a otimização emerge não como uma coleção de técnicas isoladas, mas como uma operação algébrica inerente à estrutura relacional. O desenvolvimento segue uma progressão incremental, partindo das relações binárias clássicas enriquecidas por quantais de Lawvere. A cada etapa é introduzido um único gerador novo e um conjunto finito de axiomas. A Álgebra Quadrática Gráfica (GQA) representa problemas de minimização da norma ℓ_2 sobre subespaços afins; a Álgebra Poliedral enriquece o arcabouço com restrições de desigualdade linear; e a Álgebra Gráfica para Programação Linear Paramétrica (GLP) constitui a contribuição central da tese, na qual um programa linear é tratado como um morfismo compositável com outros blocos de otimização. Para cada álgebra são demonstrados resultados de solidez e completude com respeito às categorias semânticas correspondentes. Um resultado central mostra que toda função convexa afim por partes admite representação como função valor de um programa linear, e que essa equivalência se manifesta diagramaticamente como uma forma normal canônica, eliminando no nível sintático a distinção entre um LP e uma função convexa. O cálculo diagramático para GLP desempenha, para a programação linear, o mesmo papel que a Álgebra Linear Gráfica desempenha para matrizes: fornece uma linguagem de formas normais, regras de reescrita e teoremas de completude dentro dos quais argumentos de otimização podem ser conduzidos simbolicamente, antes e independentemente de qualquer computação numérica.

Palavras-chave: teoria das categorias; diagramas de corda; programação linear; otimização convexa; categorias monoidais simétricas; relações ponderadas; quantais; álgebra gráfica; formas normais; completude.

ABSTRACT

This dissertation investigates how optimization problems can be understood within a unified compositional and diagrammatic framework grounded in category theory. The central observation is that key mathematical structures in optimization — relations, convex functions, quadratic forms, polyhedra, and linear programs — can be interpreted as morphisms in categories whose composition corresponds to the elimination or minimization of internal variables. From this perspective, optimization emerges not as a collection of isolated techniques but as an algebraic operation intrinsic to relational structure. The development follows an incremental progression rooted in Lawvere quantales and weighted relations, adding one new generator and a finite set of axioms at each step. Graphical Quadratic Algebra (GQA) expresses minimization of the ℓ_2 norm over affine subspaces; Polyhedral Algebra enriches the framework with linear inequality constraints; and Graphical Linear Programming (GLP) — the central contribution — treats a linear program as a morphism that composes with other optimization blocks, so that the elimination of internal variables corresponds exactly to categorical composition. Soundness and completeness results are proved for each algebra with respect to their semantic categories of cost relations and polyhedral relations. A key result shows that every piecewise affine convex function can be represented as the value function of a linear program, and that this equivalence manifests diagrammatically as a canonical normal form, erasing the syntactic distinction between an LP and a convex function. The diagrammatic calculus established for GLP plays the same role for linear programming that Graphical Linear Algebra plays for matrices: it provides a language of normal forms, rewriting rules, and completeness theorems within which optimization arguments can be carried out symbolically, prior to and independently of numerical computation.

Keywords: category theory; string diagrams; linear programming; convex optimization; symmetric monoidal categories; weighted relations; quantales; graphical algebra; normal forms; completeness.

SUMÁRIO

1 INTRODUCTION

Mathematics is permeated by this notion of what one would call an ”ideal” object. This notion of ”ideal” often refers to the simplest object that meets a specific criteria, such as being the smallest element in a set, or the one that factor out all others. This particular description of ”ideal”, reveals a interest interconnection between categories and optimization. Consider, for instance, the greatest common divisor of two integers $a, b \in \mathbb{Z}$. This is, from an optimization perspective, to meet the max object from the set of common divisor of a, b, categorically, this is precisely the terminal object of the poset of common divisors of a and b ordered by divisibility. Prior works have already characterised the internal optimization structure within categorical constructs [Kad26], but such results are still isolated.

This thesis is motivated by this observation: We believe that there's a very expressive and surprisingly suitable optimization view to mathematics, and that many important structures can actually be described as the solution of a particular optimization problem. In an analogous manner, we think that, just as some seminal works have tried to argue [Fre79; WR10], mathematics, and therefore optimization, can be understood within a relational framework. By relation, we mean any logical proposition of the form xRy. This notion sits within the set-theoretic foundation of mathematics, under the interpretation of a subset $S \subset X \times Y$, which is, in turn, a generalization of maps between spaces.

Category theory is, in turn, a natural fit for such a view, given that one of the main results of the field is the Yonneda Lemma, which states that an object is fully grasped by the relations it has to others [Lan98]. More interestingly, we can actually talk about the optimal object from two different perspectives, namely: a categorical one through the lens of terminal/initial objects and an optimization one through the ideas of ordered and measure, which we ought to show that, in the end, are one and the same.

Formally, we show that under specific formulation, solving an optimization problem is equivalent of performing algebraic composition in a suitable category, which generalize other notions such as variable elimination and, therefore, relational composition. More broadly, that convex optimization can actually be constructed as a single compositional framework [SS24; Han+24]. In this way, optimization itself becomes an algebraic process and an internal setting across mathematics.

This expressivity becomes especially apparent when analyzed through the language of string diagrams [JS91; Sel10; PZ23]. Originating as a rigorous graphical language for monoidal categories, string diagrams provide an intuitive yet mathematically precise syntax in which the sequential and parallel composition of morphisms correspond directly to vertical and horizontal juxtaposition of diagrams (e.g. computational processes) [JS91; Lan98]. Building on this foundation, a rich family of *diagrammatic* or *graphical calculi* has emerged, offering complete axiomatisations of various algebraic structures purely through equational diagram rewriting [Sel10; BSZ17]. These calculi have proved remarkably versatile, finding applications across a wide range of fields: in *quantum computing* [CD11]; *electronics* [BF18]; *linear algebra* [BSZ17; Bon+19; Fre+25]; and in *optimisation and probability* [Ste+24; SS21; BDS21]. Throughout this work we explore how, step by step, one can enrich the already known theories for linear [BSZ17] algebra by adding one new generator at a time, along with a particular set of axioms, and derive even richer structures for convex problems [SS24].

The development followed a strict linear discipline, adding exactly one new generator at a time and verifying at each step that the resulting algebra is sound, expressive, and complete with respect to a precisely defined semantic category. This methodology is itself one of the work's implicit arguments. The fact that such a linear development is possible, that each new class of optimization problems requires only a single additional generator and a finite set of axioms, is evidence that the compositional structure is not imposed artificially but is genuinely present in the mathematics. This is made precise by the normal form and completeness results of Theorem 3.3 and Theorem 4.1, which follow the very same pattern already established for graphical linear and affine algebra [BSZ17; Bon+19] and for their quadratic [Ste+24] and polyhedral [BDS21] extensions.

We start by extending the Symmetric Monoidal Theory of Graphical Affine Algebra (**GAA**) [Bon+19] into the even more general theory of Symmetric Inequality Monoidal Theory. This is achieved by adding the inequality generator $- \geq -$. Geometrically, this allows our algebra to go beyond affine spaces and start representing polyhedral figures as linear inequality constraints in optimization.

$$\{|x \in P\} = \begin{cases} 0 & if\ Ax + b \geq 0 \\ +\infty & \text{otherwise} \end{cases}$$

Inspired by these observations, and parallel works on this [Ste+24; Law73] we make our central contribution in the final section. The introduction of graphical algebra for parametric linear programming. In this setting, a linear program is no longer viewed only as a numerical problem to be solved, but as a morphism that can be composed with other optimization blocks. This gives a symbolic language in which the elimination of internal variables corresponds directly to categorical composition, allowing optimization problems to be manipulated diagrammatically before computation.

In particular, every piecewise affine convex function can be represented as the value function of a linear program [Roc70], which means the Graphical Linear Programming **GLP** theory provides a syntax for a broad class of nonsmooth convex models that appear in optimization, control, economics, and machine learning. In Section 4 we show the diagrammatical equivalence of this two classes of objects and how diagrammatically they are instances of the same normal form. Thus, our diagrammatical syntax blurs the distinction between expressing LP programs as optimization problems and piecewise convex functions. In other words, more poetically stated, *our diagrammatical theory unifies under the same view, the relational and optimization nature of linear programming.*

More broadly, the thesis suggests that graphical methods can play for optimization the same role that graphical linear algebra plays for matrices: providing normal forms, symbolic rewriting rules, and completeness results within a rigorous categorical setting. This reinforces the effort of using diagrammatic reasoning not only for proofs, but also for designing compositional optimization systems and formal symbolic calculi for convex programming.

Thesis overview In Section 2.1 we introduce the algebraic and categorical playground on which the rest of the work is built, present the initial notions of quantales and relations, and how they play an interchangeable role in a categorical view. Next, in Section 2.2, we introduce a basic application of this theory to the vast field of convex optimization [SS24; Kad26], showing how many of its important constructs fall within such a categorical framework.

Chapter 3 presents the already-known graphical theory of Polyhedral Algebra [BDS21], where some of the known proofs are recalled and refined.

Finally, in Chapter 4, we define and prove the completeness of an entirely new Graphical Algebra dedicated to expressing parametric Linear Programs. This work represents a first step toward a complete diagrammatic calculus for convex optimization.

2 BACKGROUND

2.1 RELATIONS AND QUANTALES

In this section we explore the theory of relations. While much of mathematical theory is built on functional thinking, we argue that most of mathematics actually has a relational nature. Category theory is the natural language for this purpose. It is precisely built on top of this notion, in which everything one needs to know about an object is determined by its morphisms.

To begin we define the most basic kind of relation:

Definition 2.1 (Binary Relations)**.** *Let X and Y be sets. A* binary relation *R from X to Y is a subset*

$$R \subseteq X \times Y.$$

Equivalently, it can be identified with its characteristic function

$$\chi_R : X \times Y \to \{0, +\infty\},$$

defined by

$$\chi_R(x, y) = \begin{cases} 0 \ \textit{if } (x, y) \in R, \\ +\infty \quad \textit{otherwise.} \end{cases}$$

A binary relation specifies whether x is related to y. Observe that one can think of a relation in two different ways that will prove useful later. First, as a set of pairs, where every relation is just a pair $(x, y) \in R$, or as a map that has x, y as inputs and outputs 0 if they are related, and $+\infty$ if they are not the choice of ∞ is arbitrary and will be justified later. Normal choices here are actually $\{true, false\}$ or $\{0, 1\}$.

This is a general construct as the sets X and Y can be anything. We can, for instance, ask for further structures on such sets and obtain different kinds of relations.

Definition 2.2 (Linear/Affine Relations)**.** *Let X and Y be vector spaces. A* linear relation *from X to Y is a linear subspace*

$$R \subseteq X \times Y.$$

An affine relation *is an affine subspace of $X \times Y$.*

Equivalently, a linear relation can be described as the graph of a possibly multivalued linear operator, i.e.,

$$R(x) := \{y \in Y \mid (x, y) \in R\}.$$

or, its characteristic function defined by

$$\chi_R : X \times Y \to \{0, +\infty\},$$

$$\chi_R(x,y) = \begin{cases} 0 \ \textit{if } (x,y) \in R, \\ +\infty \quad \textit{otherwise.} \end{cases}$$

Thus, a linear/affine relation is a binary relation where the set R is a subspace

Notice, however, that we have been restricting the relations between x and y to be a case of "they are, or they are not related". In fact, we might be interested in adding degrees to such a relation, so that we could say that x and y are "more" or "less" related. This leads us to the well-known notion of weighted relations [Law73].

Definition 2.3 (Weighted Relations). *Let X and Y be any two sets and let $\overline{\mathbb{R}}^+ := [0, +\infty]$. A* weighted relation *from X to Y is a set*

$$R \subseteq X \times Y \times \overline{\mathbb{R}}^+$$

such that for every $(x,y) \in X \times Y$ there exists exactly one *$r \in \overline{\mathbb{R}}^+$ satisfying*

$$(x,y,r) \in R.$$

Equivalently, we may define the characteristic function

$$\chi_R : X \times Y \to \overline{\mathbb{R}}^+$$

by

$$\chi_R(x,y) = r \quad \textit{if and only if } (x,y,r) \in R.$$

and the codomain is not restricted to a binary set anymore. In fact, one could choose any other set instead of the $\overline{\mathbb{R}}^+$ to be the codomain of the characteristic function as long as it satisfy some properties. In fact, the whole structure of a relation is further generalized by the notion of a quantale.

Definition 2.4 (Quantale). *A* quantale *is a structure $(Q, \leq, \otimes, e)$ such that:*

- *$(Q, \leq)$ is a complete lattice, i.e., every subset $A \subseteq Q$ admits a join $\bigvee A$ and a meet $\bigwedge A$;*
- *$(Q, \otimes, e)$ is a monoid;*
- *Multiplication distributes over arbitrary join in each variable:*

$$a \otimes \Big(\bigvee_{i \in I} b_i\Big) = \bigvee_{i \in I} (a \otimes b_i), \qquad \Big(\bigvee_{i \in I} a_i\Big) \otimes b = \bigvee_{i \in I} (a_i \otimes b)$$

  for all families $\{a_i\}_{i \in I}, \{b_i\}_{i \in I} \subseteq Q$.

If $\otimes$ is commutative, the quantale is called commutative*. If $e = \top$ (the top element of the lattice), it is called* integral*.*

Given a quantale Q, we can define the notion of Q-valued relation [Coe+18].

Definition 2.5 (Generalized Relations)**.** *A* generalized relation *(or Q-valued relation) from X to Y, where Q is a quantale, is a set*

$$R \subseteq X \times Y \times Q$$

such that for every $(x, y) \in X \times Y$ there exists exactly one *$q \in Q$ satisfying*

$$(x, y, q) \in R.$$

Its characteristic function is

$$\chi_R : X \times Y \to Q$$

defined by

$$\chi_R(x, y) = q \quad \textit{if and only if } (x, y, q) \in R.$$

Now we are ready to build our previous relations as particular cases of this more general framework.

Example 2.1. *The notion of a Q-valued relation recovers familiar concepts for particular choices of quantale.*

(1) Truth quantale. *Let $Q = \{0, 1\}$ with the usual order $0 \leq 1$, join given by $\vee$, and monoidal product $\otimes = \wedge$ with unit $e = 1$. This is the (Boolean) truth quantale.*

A Q-valued relation is then a function

$$R : X \times Y \to \{0, 1\}.$$

Such a function is precisely the characteristic function of an ordinary binary relation

$$\widetilde{R} \subseteq X \times Y, \qquad (x, y) \in \widetilde{R} \iff R(x, y) = 1.$$

Thus Q-valued relations coincide with classical binary relations.

(2) Lawvere quantale. *Let $Q = \overline{\mathbb{R}}^+ = [0, +\infty]$ ordered by the* reverse *of the usual order (so that $a \leq b$ iff $a \geq b$ in the usual order), with monoidal product given by addition*

$$a \otimes b := a + b,$$

and unit $e = 0$. Joins are then given by infima in the usual order.

This structure is the Lawvere quantale. A Q-valued relation

$$R : X \times Y \to \overline{\mathbb{R}}^{+}$$

is precisely a weighted relation assigning a (possibly infinite) cost to each pair (x, y).

Hence generalized relations unify ordinary binary relations and weighted relations as special cases corresponding to particular quantales.

Note that linear and affine relations are not recovered by this construction, since they demand extra structure on the sets X and Y, specifically that R be a linear or affine subspace of $X \times Y$. They are, nonetheless, fully embedded in the framework of binary relations.

Within this framework, we propose an extension of the Lawvere Quantale and weighted relations called the Extended-Lawvere Quantale and the Cost/Reward Relations, respectively. To our knowledge, this definition does not appear elsewhere in the literature.

Definition 2.6 (Optimization/Extended Lawvere Quantale)**.** *Let*

$$\overline{\mathbb{R}} := \mathbb{R} \cup \{-\infty, +\infty\}.$$

The optimization quantale *is the structure*

$$(\overline{\mathbb{R}}, \geq, \otimes, e)$$

defined as follows:

- *The order $\geq$ is the usual order on the extended real line. With this order, arbitrary joins are given by the usual infima:*

$$\bigvee A = \inf A.$$

- *The monoidal product is addition:*

$$a \otimes b := a + b,$$

where addition is extended in the usual way (with $a+(+\infty) = +\infty$ and $a+(-\infty) = -\infty$ whenever defined).

- *The unit element is*

$$e := 0.$$

With these operations, $(\overline{\mathbb{R}}, \geq, +, 0)$ is a commutative quantale. Joins are ordinary infima, and addition distributes over arbitrary joins.

Definition 2.7 (Optimization (Cost/Reward) Relations)**.** *Let X and Y be sets. An* optimization relation *from X to Y is an $\overline{\mathbb{R}}$-valued relation, i.e., a function*

$$R : X \times Y \to \overline{\mathbb{R}}.$$

Equivalently, it is a subset

$$R \subseteq X \times Y \times \overline{\mathbb{R}}$$

such that for every $(x, y) \in X \times Y$ there exists exactly one $r \in \overline{\mathbb{R}}$ satisfying

$$(x, y, r) \in R.$$

The value $R(x, y)$ is interpreted as the objective value (cost or reward) associated to the pair (x, y), with $+\infty$ and $-\infty$ representing extremal infeasibility or unboundedness.

Observe how this is essentially the weighted relations extended along the negative values. The importance of it is that we want to be able to express optimization problems that might be unfeasible ($+\infty$) or unbounded ($-\infty$), so we need to add both the top and bottom element.

All of this admits a suitable and natural categorical construction. Specifically, from any quantale Q one can not only define Q-relations, but also construct an entire category of them. This category is the natural setting for discussing the central concept of this framework: the composition of relations.

The following lemma shows how to build such a category and establishes that this category also has a very important monoidal structure behind it. Particularly, the category is a compact closed symmetric monoidal category [Lan98]:

Proposition 2.1 (Category $\mathbf{Rel}(Q)$)**.** *For a quantale Q, the Q-relations form a category* ***Rel***_(Q)_ *with composition and identities*

$$(S \circ R)(a, c) = \bigvee_b R(a, b) \otimes S(b, c) \quad 1_A(a, b) = \bigvee \{k | a = b\}$$

If Q is a commutative quantale, then ***Rel***_(Q)_ *carries a symmetric monoidal structure, with the categorical monoidal product $\bigotimes_{\mathcal{C}}$ on objects given by the Cartesian product of sets and the action on relations given for $R : A \to C$ and $S : B \to D$ by*

$$(R \otimes_{\mathcal{C}} S)(a, b, c, d) = R(a, c) \otimes S(b, d)$$

The singleton set is the monoidal unit, and the category is compact closed w.r.t the monoidal structure

Proof. We verify the categorical structure.

(1) Well-defined composition. Let $R : A \to B$ and $S : B \to C$ be Q-relations. For each $(a, c) \in A \times C$, the set

$$\{R(a, b) \otimes S(b, c) \mid b \in B\} \subseteq Q$$

admits a supremum since Q is a complete lattice. Hence $(S \circ R)(a, c)$ is well-defined as an element of Q.

(2) Associativity. Let $R : A \to B$, $S : B \to C$, and $T : C \to D$. For $a \in A$, $d \in D$,

$$((T \circ S) \circ R)(a, d) = \bigvee_c \Big(\bigvee_b R(a, b) \otimes S(b, c) \Big) \otimes T(c, d).$$

Since $\otimes$ distributes over arbitrary suprema,

$$= \bigvee_{b,c} R(a, b) \otimes S(b, c) \otimes T(c, d).$$

Similarly,

$$(T \circ (S \circ R))(a, d) = \bigvee_b R(a, b) \otimes \Big(\bigvee_c S(b, c) \otimes T(c, d) \Big) = \bigvee_{b,c} R(a, b) \otimes S(b, c) \otimes T(c, d).$$

Thus composition is associative.

(3) Identities. For each set A, define

$$1_A(a, b) = \begin{cases} e & \text{if } a = b, \\ \perp & \text{otherwise,} \end{cases}$$

where $\perp = \bigvee \varnothing$ is the bottom element of Q.

Then for $R : A \to B$,

$$(R \circ 1_A)(a, b) = \bigvee_{a'} 1_A(a, a') \otimes R(a', b).$$

All terms vanish except when $a' = a$, so

$$= e \otimes R(a, b) = R(a, b).$$

Similarly, $1_B \circ R = R$. Hence identities exist.

(4) Monoidal structure (commutative case). Assume Q is commutative. Define the tensor on objects by Cartesian product:

$$A \otimes_{\mathcal{C}} B := A \times B.$$

For relations $R : A \to C$ and $S : B \to D$, define

$$(R \otimes_{\mathcal{C}} S)((a,b),(c,d)) = R(a,c) \otimes S(b,d).$$

Functoriality follows from associativity and commutativity of $\otimes$:

$$(S_1 \otimes S_2) \circ (R_1 \otimes R_2) = (S_1 \circ R_1) \otimes (S_2 \circ R_2).$$

The singleton set $\mathbf{1}$ is a unit since $A \times 1 \cong A$.

Symmetry is induced by the swap map

$$\sigma_{A,B}((a,b),(b',a')) = \begin{cases} e & \text{if } a = a',\ b = b', \\ \perp & \text{otherwise.} \end{cases}$$

(5) Compact closure. For each set A, define unit and counit relations

$$\eta : 1 \to A \times A, \qquad \epsilon : A \times A \to 1$$

by

$$\eta(*,(a,a')) = \begin{cases} e & a = a', \\ \perp & \text{otherwise,} \end{cases} \qquad \epsilon((a,a'),*) = \begin{cases} e & a = a', \\ \perp & \text{otherwise.} \end{cases}$$

A direct computation using the definition of composition shows that the triangle identities hold:

$$(1_A \otimes \epsilon) \circ (\eta \otimes 1_A) = 1_A, \qquad (\epsilon \otimes 1_A) \circ (1_A \otimes \eta) = 1_A.$$

Thus every object is self-dual, and $\mathbf{Rel}(Q)$ is compact closed. □

A curious fact about relations is that they actually encode the notion of matrices and matrix multiplication. This serves as motivation for what will later become the diagrammatical theory of linear algebra. The following remark expands on it showing that every morphism in $\mathbf{Rel}(Q)$ corresponds to a matrix with entries in Q, and that composition is exactly matrix multiplication.

Remark 2.1. *A Q-valued relation $R : X \times Y \to Q$ is nothing but a Q-*valued matrix *indexed by $X \times Y$: one simply reads off the entry at position (x,y) as $\chi_R(x,y) \in Q$. That is, a generalized relation is precisely an element of $Q^{X \times Y}$.*

When X and Y are finite sets, say $|X| = m$ and $|Y| = n$, this becomes completely explicit: a Q-valued relation is exactly an $m \times n$ matrix

$$M_R \;=\; \big[\chi_R(x_i,\, y_j)\big]_{\substack{1 \le i \le m \\ 1 \le j \le n}} \;\in\; Q^{m \times n},$$

whose (i, j)-entry records the Q-value assigned to the pair (x_i, y_j).

Moreover, composition of Q-valued relations corresponds precisely to matrix multiplication in Q. *Given a second Q-valued relation $S : Y \times Z \to Q$, their composite $S \circ R : X \times Z \to Q$ is defined by*

$$\chi_{S \circ R}(x, z) \;=\; \bigvee_{y \in Y} \chi_R(x, y) \otimes \chi_S(y, z),$$

which, in matrix notation, reads

$$M_{S \circ R} \;=\; M_R \otimes M_S, \qquad (M_R \otimes M_S)_{ij} \;=\; \bigvee_{k=1}^{n} (M_R)_{ik} \otimes (M_S)_{kj}.$$

This is exactly the usual row-times-column matrix product, but with ordinary addition replaced by the join $\bigvee$ and ordinary multiplication replaced by the monoidal product $\otimes$ of Q. The two axioms of a quantale — that $\otimes$ distributes over arbitrary joins — are precisely what guarantees that this multiplication is associative and well-defined, so that Q-valued matrices form a category under this composition.

To illustrate these new concepts, we look at some of our previous relations now built as categories.

Example 2.2 (Category of Boolean relations)**.** *Let*

$$Q = (\{0, 1\}, \leq, \otimes, e)$$

be the Boolean quantale where:

- $0 \leq 1$*;*
- $\otimes = \wedge$ *(logical conjunction);*
- $e = 1$*;*
- *joins are given by* $\vee$*.*

A Q-valued relation $R : A \to B$ is a function

$$R : A \times B \to \{0, 1\},$$

which is precisely the characteristic function of a subset

$$\widetilde{R} \subseteq A \times B.$$

The composition formula in $\mathbf{Rel}(Q)$ *becomes*

$$(S \circ R)(a, c) = \bigvee_{b} R(a, b) \wedge S(b, c).$$

Since joins are logical disjunction, this reduces to

$$(S \circ R)(a,c) = 1 \quad \Longleftrightarrow \quad \exists b \in B \text{ such that } R(a,b) = 1 \text{ and } S(b,c) = 1.$$

Thus composition coincides exactly with the usual composition of binary relations.

Since the Boolean quantale is commutative, $\mathbf{Rel}(Q)$ *carries a symmetric monoidal structure:*

- *On objects:*
$$A \otimes B := A \times B.$$
- *On morphisms:*
$$(R \otimes S)((a,b),(c,d)) = R(a,c) \wedge S(b,d).$$

The singleton set is the monoidal unit. With this structure, $\mathbf{Rel}(Q)$ *is precisely the classical symmetric monoidal compact closed category* **Rel***.*

Example 2.3 (Category of weighted relations)**.** *Let*

$$Q = (\mathbb{R} \cup \{+\infty\}, \leq_Q, \otimes, e)$$

be the Lawvere quantale where:

- $a \leq_Q b \iff a \geq b$ *in the usual order;*
- $\otimes = +$*;*
- $e = 0$*;*
- *joins are given by infima in the usual order.*

A Q*-valued relation* $R : A \to B$ *is a function*

$$R : A \times B \to \mathbb{R} \cup \{+\infty\},$$

assigning a cost (or weight) to each pair (a,b)*.*

The composition formula becomes

$$(S \circ R)(a,c) = \inf_{b \in B} \big(R(a,b) + S(b,c)\big).$$

Thus composition is given by infimal convolution *(or shortest-path composition).*

Since addition is commutative, the Lawvere quantale is commutative, and $\mathbf{Rel}(Q)$ *carries a symmetric monoidal structure:*

- *On objects:*
$$A \otimes B := A \times B.$$

- *On morphisms:*
$$(R \otimes S)((a,b),(c,d)) = R(a,c) + S(b,d).$$

The singleton set is the monoidal unit. With this tensor product, $\mathbf{Rel}(Q)$ *is a symmetric monoidal compact closed category of weighted relations.*

In particular:

- *If* A, B, C *are finite sets, relations may be viewed as matrices with entries in* $\mathbb{R} \cup \{+\infty\}$.

- *Composition is matrix multiplication in the* $(\min, +)$*-semiring.*

Hence $\mathbf{Rel}(Q)$ *is the category of weighted relations, which underlies dynamic programming, tropical algebra, and shortest-path problems.*

Those examples illustrate how naturally the quantale structure supports categorical constructions: composition of Q-valued relations corresponds to matrix multiplication driven by $\otimes$, and the tensor product of relations is governed by the join $\bigvee$, both operations coming directly from the quantale axioms.

The resulting categories are symmetric monoidal, and it is precisely this monoidal backbone that will support the algebraic structure we aim to build. A key piece of that structure is a *Frobenius monoid* on each object: a monoid (μ, η) and a comonoid (δ, ε) on the same object, satisfying the Frobenius law

$$(\mathrm{id} \otimes \mu) \circ (\delta \otimes \mathrm{id}) \;=\; \delta \circ \mu \;=\; (\mu \otimes \mathrm{id}) \circ (\mathrm{id} \otimes \delta).$$

A symmetric monoidal category in which every object carries such a structure, compatibly with the tensor product, is called a *hypergraph category* [FS19].

Hypergraph categories are related to, but strictly more general than, compact closed categories [Sel10]. In a compact closed category every object A has a dual A^* with unit $\eta_A : I \to A^* \otimes A$ and counit $\varepsilon_A : A \otimes A^* \to I$ satisfying the snake equations; a Frobenius structure does not require the existence of such a dual. Conversely, if a compact closed category is such that every object is self-dual ($A \cong A^*$) and the induced cups and caps satisfy the Frobenius law, then it is a hypergraph category. Thus compact closed categories with self-dual objects form a special class of hypergraph categories, and it is the later, more general, notion that we shall adopt as our ambient framework.

Remark 2.2. *If Q is commutative, then* $\mathbf{Rel}(Q)$ *is a hypergraph category and admits a commutative Frobenius structure on every object. Diagrammatically, we denote the additive and co-additive structure as*

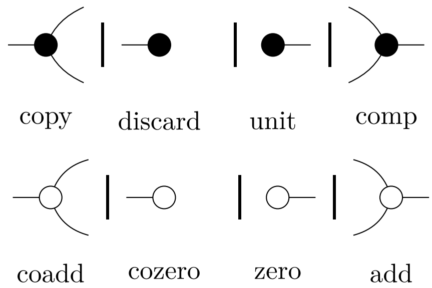
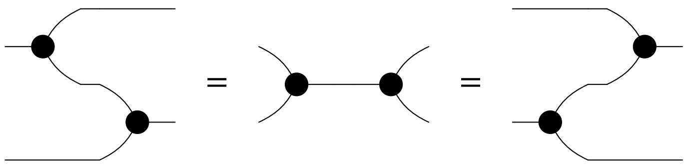


Figure 2.1 – Frobenius additive(black) and co-additive(white) structure

Subject to the Frobenius Law (both colors)



Proof. Just a check that all laws do hold for any $\mathbf{Rel}(Q)$ category $\square$

These constructions provide exactly the semantic foundation for the algebra we aim to build. In the same way that semantics is understood in linguistics and programming language theory [Win93], the semantics of a formal system assigns meaning to its expressions. More precisely, while syntax consists of the tokens and rules for forming and combining expressions, semantics is the interpretation assigned to each such expression — in our case, an optimization problem living in $\mathbf{Rel}(Q)$.

As outlined in the introduction, the central goal of this work is to show that optimization emerges naturally from algebraic and categorical structure. The key mechanism enabling this is the infimal composition present in the definition of the Costs/Rewards category: composing two relations automatically performs variable elimination, which is precisely the fundamental operation underlying optimization.

To make this precise, we need both a semantics — provided by the categorical framework developed above — and a syntax: a rigorous, compositional language for expressing optimization problems symbolically. The map between those is structure-preserving map, a functor, from the syntactic layer into the semantic one.

The syntactic layer will be formalized as a Symmetric Monoidal Theory (SMT) [BSZ17], which is a presentation of a category by generators and equations, whose terms respect the compositional structure of a symmetric monoidal category. This is the standard tool for building graphical languages in the style of Graphical Linear Algebra, and it is the framework within which our completeness results will be stated and proved.

Definition 2.8 (Symmetric Monoidal Theory)**.** *A SMT is a pair* (Σ, E)*, such that,* Σ *is a signature of operators* o *that has arity* m *and coarity* n

Σ-terms are defined inductively as the smallest collection of typed expressions $t : m \to n$ such that:

1. *(Generators) If $o \in \Sigma$ with $o : m \to n$, then o is a Σ-term of type $m \to n$.*
2. *(Identity) For every $n \in \mathbb{N}$, there is a term*
$$\mathrm{id}_n : n \to n.$$
3. *(Composition) If*
$$a : m \to n \quad \textit{and} \quad b : n \to k$$
are Σ-terms, then
$$a; b : m \to k$$
is a Σ-term.
4. *(Tensor product) If*
$$a : m_1 \to n_1 \quad \textit{and} \quad b : m_2 \to n_2$$
are Σ-terms, then
$$a \otimes b : m_1 + m_2 \to n_1 + n_2$$
is a Σ-term.
5. *(Symmetry) For all $m, n \in \mathbb{N}$, there is a term*
$$\sigma_{m,n} : m + n \to n + m.$$

and E be a set of equations over the $\Sigma - terms$.

Often, we will think of elements $o : m \to n \in \Sigma$ as generators and represent them as string diagrams. Being the $\Sigma - terms$ the results of composing those generators horizontally and vertically inductively. The set E as a collection of axiomatic equations over these string diagrams.

The categorical counterpart of these terms is the free category generated by the SMT.

Definition 2.9 (The Free category of SMT)**.** *The $FreeP_{(\Sigma,E)}$ is the category freely generated by (Σ, E) where the objects are natural numbers, morphisms are $\Sigma - terms$ quotiented by E, with composition and tensor product inherited from the symmetric monoidal structure. $FreeP_{(\Sigma,E)}$ is in fact a PROP*[1]*.*

[1]For the definition of a prop see [Lan98]

The $\mathrm{FreeP}_{(\Sigma,E)}$, also denoted as $P\Sigma_{/E}$, [2] will be our **syntax** responsible for expressing our underlying semantics. For it to be considered a "useful" syntax we expect that it satisfies some properties, particularly we want it to be complete, sound and expressive.

Without those, our algebra would be empty since we would have no assurances that it actually says anything meaningful about we desire to say.

Definition 2.10. *We will say that a SMT axiomatizes or represents a specific category, usually called the semantic category, when the following diagram commutes:*

$$\begin{tikzcd}
\mathbf{P\Sigma}_{/E} = FreeP_{(\Sigma,E)} \arrow[rr, "\cong"] & & \mathrm{Im}(\llbracket\cdot\rrbracket) \arrow[d, hook, "i"] \\
\mathbf{Syn} = \mathbf{P\Sigma} \arrow[u, "q"] \arrow[urr, "s"] \arrow[rr, "\llbracket\cdot\rrbracket"'] & & \mathbf{Sem}
\end{tikzcd}$$

*where: $P\Sigma$ is the category where objects are natural numbers and morphisms are Σ***−terms** *from $n \to m$; $P\Sigma_{/E} = FreeP_{(\Sigma,E)}$ is the $P\Sigma$ category with the morphisms quotiented by E and there is then a prop morphism $q : P\Sigma \to P\Sigma_{/E}$ witnessing this quotient;* ***Sem*** *is our semantic category (the Cost/Reward relations for example); and s is a full and faithful prop morphism. Another way to say the same is to demand two things out of the SMT, namely:*

- *Soundness: We say that $P\Sigma_{/E}$ is sound if $c \overset{E}{=} d$[3] implies $[c] = [d]$.*
- *Completeness: We say that $P\Sigma_{/E}$ is complete if $[c] = [d]$ implies $c \overset{E}{=} d$*
- *Expressivity: We say that $P\Sigma_{/E}$ is expressive if for every morphism c in* ***Sem*** *there exists $d \in P\Sigma_{/E}$ such that $[d] = c$*

Those three properties capture the idea that everything we can prove in the syntax, is a holds in the semantics (soundness), everything that holds in the semantics is provable in the syntax (completeness) and that every morphism in the semantics has a syntactic representative (expressivity). Moreover, these properties denote equivalent characteristics about the interpration map, both categorically and functionally. Below we depict a small table defining the equivalence between the desired properties and their equivalent statement:

Now we introduce two of the foundational theories for Graphical Algebra, that will serve as a foundation for the following sections, namely, Graphical Linear Algebra [PRS22;

[2] During the work we will use FreeP_S and S, for whatever syntax S may be, interchangeably.

[3] $\overset{E}{=}$ means that this equality is provable by the equations in E

Table 2.1 – Equivalence of properties for the interpretation map $[\![-]\!] : P\Sigma_{/E} \to \mathbf{Sem}$

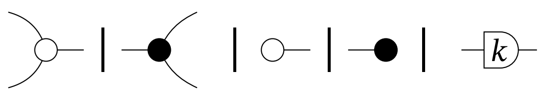
Set theory	**Category theory**	**Logic (SMT)**
Total	Functor	Soundness: $c \overset{E}{=} d \Rightarrow [\![c]\!] = [\![d]\!]$
Deterministic	Well-defined map	(Functionality of the interpretation)
Injective	Faithful	Completeness: $[\![c]\!] = [\![d]\!] \Rightarrow c \overset{E}{=} d$
Surjective	Full	Expressivity: $\forall f \in \mathbf{Sem}, \exists d \in P\Sigma_{/E}$ s.t. $[\![d]\!] = f$

BSZ17] and Graphical Affine Algebra [Bon+19], we omit their definitions and proofs and refer to the original sources.

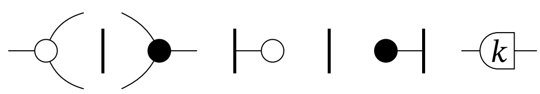
However, we do provide some notation on how to talk about those during the course of the work. Notice that, their semantics was already introduced when we talked about Linear/Affine relations.

Definition 2.11. *We will call $\boldsymbol{GLA}$ the presentation of the Graphical Linear Algebra SMT, and, in the same manner, $\boldsymbol{GAA}$ the SMT of Graphical Affine Algebra.*

Definition 2.12. *We also denote with $\overrightarrow{(\cdot)}/\overleftarrow{(\cdot)}$ when we want to talk about the algebra generated without the Frobenius structure. For example $\overrightarrow{\boldsymbol{GLA}}$ is the algebra generated by*



without their symmetric



and the exactly opposite we denote as $\overleftarrow{\boldsymbol{GLA}}$.

2.2 COMPOSABILITY OF CONVEX PROBLEMS

Having established the necessary categorical and algebraic tools, we proceed to their application in convex analysis. In this section we introduce an initial categorical framework for convex optimization [SS24], showing how the fundamental operations of convex analysis

— infimal convolution, conjugation, and variable elimination — arise naturally as categorical constructions. This gives the first concrete evidence that optimization is not merely a collection of isolated techniques, but a coherent algebraic theory with deep compositional structure.

The progression of the section is as follows. We begin with the general category of bifunctions, which serves as the broadest setting, and progressively restrict to convex bifunctions, where the analytical structure is richest. We then introduce the Legendre–Fenchel transform and show that bifunction adjunction defines a contravariant functor between the convex and concave bifunction categories. These results lay the groundwork for the graphical theories developed in the following sections: Graphical Quadratic Algebra and Polyhedral Algebra will both arise as full subcategories of **CxBiFn**, each corresponding to a class of bifunctions with additional algebraic structure.

Later, we will see the particular applications of such definitions and results. For now, however, we begin by defining the broadest category in the hierarchy, which will generalize most of the later work: the category of bifunctions.

Definition 2.13 (Bifunction Category)**.** *The category* **BiFunc** *is defined as follows:*

- *Objects are sets.*
- *A morphism $F : A \to B$ is a function*

$$F : A \times B \to \overline{\mathbb{R}}.$$

- *Given $F : A \to B$ and $G : B \to C$, their composition $G \circ F : A \to C$ is defined pointwise by*

$$(G \circ F)(a, c) := \inf_{b \in B} \big(F(a, b) + G(b, c)\big).$$

- *The identity morphism on an object A is the function*

$$1_A : A \times A \to \overline{\mathbb{R}}$$

given by

$$1_A(a, a') = \begin{cases} 0 & \textit{if } a = a', \\ +\infty & \textit{otherwise.} \end{cases}$$

If we equip **BiFunc** *with a monoidal structure:*

- *The tensor product on objects is the Cartesian product*

$$A \otimes B := A \times B.$$

- *For morphisms* $F : A \to C$ *and* $G : B \to D$*, their tensor product*

$$F \otimes G : A \times B \to C \times D$$

is defined by

$$(F \otimes G)((a,b),(c,d)) := F(a,c) + G(b,d).$$

The singleton set serves as the monoidal unit.

Remark 2.3. *The category* **BiFunc** *is isomorphic to the category* ***RelQ*** *for a suitable choice of* Q.

Consider the quantale

$$Q = (\overline{\mathbb{R}}, \leq_Q, \otimes, e)$$

where:

- $\overline{\mathbb{R}} = \mathbb{R} \cup \{-\infty, +\infty\}$*;*
- *the order is reversed with respect to the usual order,*

$$a \leq_Q b \quad \Longleftrightarrow \quad a \geq b \quad \textit{(usual order)};$$

- *the monoidal product is addition,*

$$a \otimes b := a + b;$$

- *the unit is* $e = 0$.

In this quantale, arbitrary joins are given by infima in the usual order:

$$\bigvee\nolimits_Q A = \inf A.$$

Now a Q*-valued relation* $R : A \to B$ *is precisely a function*

$$R : A \times B \to \overline{\mathbb{R}},$$

i.e. a bifunction. Moreover, the composition formula in $\mathbf{Rel}(Q)$ *becomes*

$$(S \circ R)(a,c) = \bigvee_b R(a,b) \otimes S(b,c) = \inf_b \big(R(a,b) + S(b,c)\big),$$

which is exactly the composition law in **BiFunc**.

Therefore, **BiFunc** *is (strictly) the same category as* $\mathbf{Rel}(Q)$ *for the Lawvere-type quantale* $(\overline{\mathbb{R}}, \geq, +, 0)$.

Bifunctions are fundamental objects in convex analysis [Roc70]. They provide a unified language for expressing objective functions, feasibility constraints, and duality simultaneously. The following proposition confirms that the framework is conservative: classical set-theoretic relations embed faithfully into this richer setting.

Proposition 2.2. *The category of Binary Relations, i.e. the Boolean-relation, is embedded as a full subcategory of* **BiFunc**.

Proof. Just consider the functor $F : \mathbf{Rel}(B) \to \mathbf{BiFunc}$ that takes every morphism $R : X \to Y$, which is a subset of $X \times Y$, and maps it into the characteristic function $\chi_R : X \times Y \to \{0, +\infty\}$, also known as the indicator function $\mathbf{1}_R(x, y)$, or in bracket notation $\{|(x, y) \in R|\}$ □

Having established the general framework, we now restrict our attention to the convex setting, which is the natural home for optimization. The key insight is that convexity is preserved under infimal composition, so the convex bifunctions form a subcategory of **BiFunc** that is closed under the categorical operations.

Definition 2.14 (Convex Bifunction Category **CxBiFn**)**.** *Let*

$$Q := (\overline{\mathbb{R}}, \leq_Q, \otimes, e)$$

be the extended-Lawvere quantale where:

- $\overline{\mathbb{R}} = \mathbb{R} \cup \{-\infty, +\infty\}$*;*
- $a \leq_Q b \iff a \geq b$ *in the usual order;*
- $a \otimes b := a + b$*;*
- $e := 0$*.*

The category **CxBiFn** *is defined as follows:*

- ***Objects:*** *finite-dimensional real vector spaces.*
- ***Morphisms:*** *A morphism $F : A \to B$ is a function*

$$F : A \times B \to \overline{\mathbb{R}}$$

satisfying:

  - *F is proper (not identically $+\infty$);*
  - *F is convex as a function on the product space $A \times B$;*
  - *F is lower semicontinuous.*

- ***Composition:*** *Given $F : A \to B$ and $G : B \to C$, their composite*

$$G \circ F : A \to C$$

is defined by infimal convolution:

$$(G \circ F)(a, c) := \inf_{b \in B} \big(F(a, b) + G(b, c)\big).$$

- ***Identity:*** *For each object A, the identity morphism*

$$1_A : A \to A$$

is given by

$$1_A(a, a') = \begin{cases} 0 & \textit{if } a = a', \\ +\infty & \textit{otherwise.} \end{cases}$$

- *Monoidal product*
  - *On objects:*

$$A \otimes B := A \times B.$$

  - *On morphisms: For $F : A \to C$ and $G : B \to D$,*

$$(F \otimes G)((a, b), (c, d)) := F(a, c) + G(b, d).$$

  - *The monoidal unit is the zero-dimensional vector space $\{0\}$.*

Remark 2.4. *One may define the analogous category* **CvBiFn** *which is the category of concave bifunctions. For doing so, one should only change the composition from inf to sup, this is:*

$$(G \circ F)(a, c) := \sup_{b \in B} \big(F(a, b) + G(b, c)\big).$$

Every result for **CxBiFn** *will hold, in an equivalent way, to the category* **CvBiFn**

Remark 2.5. *Every finite-dimensional real vector space V is linearly isomorphic to $\mathbb{R}^n$, where $n = \dim V$.*

Hence, one may restrict the objects of **CxBiFn** *to the family*

$$\mathrm{Ob}(\mathbf{CxBiFn}) = \{\mathbb{R}^n \mid n \in \mathbb{N}\}.$$

With this choice, **CxBiFn** *becomes a skeletal category, since $\mathbb{R}^n \cong \mathbb{R}^m$ implies $n = m$.*

Also, if one considers the isomorphism $n \leftrightarrow \mathbb{R}^n$*,* **CxBiFn** *becomes a prop, this can be made strict by defining the monoidal product on objects as sum.*

$$m \otimes n = m + n$$

which is isomorphic to

$$\mathbb{R}^m \otimes \mathbb{R}^m = \mathbb{R}^m \times \mathbb{R}^m = \mathbb{R}^{m+n}$$

Observe that we have constructed a category whose morphisms are interpreted as convex optimization problems. Since every bifunction $F : \mathbb{R}^m \times \mathbb{R}^n \to \overline{\mathbb{R}}$ can be written as

$$F(x, y) = \inf_z z \quad s.t\ z = F(x, y)$$

This observation is central to the thesis. It shows that there's some sort of duality between relations and optimization. For a single morphism, however, this remains imprecise. The compositional structure yields more substantive insights.

$$(S \circ R)(x, z) = \inf_y \{R(x, y) + S(y, z)\}$$

Which shows that relational composition is precisely variable elimination: composing two relations automatically marginalises over the intermediate variable y by minimising the combined cost. In the binary case this reduces to the existential question of whether a mediating element exists; in the weighted case it asks for the *optimal* such element. Optimization is therefore not imposed on the categorical structure from outside — it emerges from composition itself.

Convex theory also has a very nice duality theory which is of particular interest in algebraic constructs such as this one. We, however, do not include such results in this thesis, leaving it for the interested reader [SS24; Wil15]

The results of this section establish **CxBiFn** as a well-behaved categorical setting for convex optimization. The special classes of morphisms of polyhedral bifunctions inherit this structure while admitting a finite, explicit representation. It is precisely this finiteness that makes a graphical algebra possible: morphisms can be encoded as diagrams composed from a finite set of generators, and the equational theory of the diagrams corresponds exactly to the equivalences between optimization problems. The following section make this precise.

3 POLYHEDRAL RELATIONS

We now turn to a combinatorially richer class of objects: polyhedra and polyhedral cones. These are the fundamental geometric structures underlying linear programming, and they admit a categorical treatment entirely analogous to the one developed for quadratic relations. The key difference is that composition here is governed by Fourier-Motzkin Elimination (FME) rather than QR factorization: projecting a polyhedron onto a subspace is precisely variable elimination, and it is this operation that categorical composition will simulate diagrammatically.

The theory developed here remains within the category of binary relations, since morphisms are still indicator functions of subsets of $\mathbb{R}^n \times \mathbb{R}^m$, encoding feasibility constraints rather than objective functions. In this sense, **GPA** does not yet enter the weighted relational world, and so is not yet able to express more meaningful optimization problems. This is nonetheless a critical step in the development. Axiomatizing linear inequality constraints diagrammatically, **GPA** provides the Symmetric Monoidal Inequality Theory (SMIT) [BDS21] that will serve as the structural foundation for our final contribution, **GLP**, where these constraints are combined with a linear objective to form full parametric linear programs.

In what follows we describe and show the complete theory of polyhedral relations and polyhedral cones. Just as **AffRel** captured affine space as morphisms, the categories **PCRel** and **PolyRel** capture polyhedral cones and polyhedra as morphisms, with relational composition playing the role of variable elimination.

3.1 SEMANTICS: THE PROP **PCREL** OF POLYHEDRAL CONES

We begin by recalling the standard definitions of polyhedral cones and the corresponding semantic category.

Definition 3.1. *A set $C \subseteq \mathbb{R}^n$ is called a* polyhedral cone *if there exists a matrix $A \in \mathbb{R}^{m\times n}$ such that*

$$C = \{x \in \mathbb{R}^n \mid Ax \geq 0\}.$$

Definition 3.2 (The prop **PCRel** of polyhedral cones)**.** *Let k be an ordered field, in this case the real numbers. The prop $\mathbf{PCRel}_k$ is defined as follows:*

- *Objects are natural numbers $n \in \mathbb{N}$, interpreted as the space k^n.*
- *A morphism $n \to m$ is a polyhedral cone $R \subseteq k^n \times k^m$ for which there exist $p \in \mathbb{N}$*

and $A \in k^{p\times(n+m)}$ *such that*

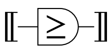
$$R = \left\{ (x,y) \in k^n \times k^m \;\middle|\; A\begin{pmatrix} x \\ \\ y \end{pmatrix} \geq 0 \right\}.$$

- *Composition is relational composition.*
- *The monoidal product on objects is addition* $n \otimes m := n + m$*, and on morphisms is the Cartesian product.*

Observe that in comparison with the previous categories of **AffRel**, this category is way more general although it does not completely encode the notion of Affine Spaces. However, it does introduce to us a somewhat unorthodox notion of conical polyhedral relations. That is, the fact that two elements x, y may be related to one another by a relation R such that the set of all pairs xRy form a polyhedral cone. Although this description remains very set-theoretically, it starts to flirt with the idea of x and y being apparted from a specific class of maps. More precisely, instead of thinking of the whole geometry of polyhedral cones, one may think of the pairs (x, y) such that $Ax \geq By$ for some A and B fully defined by the relation R.

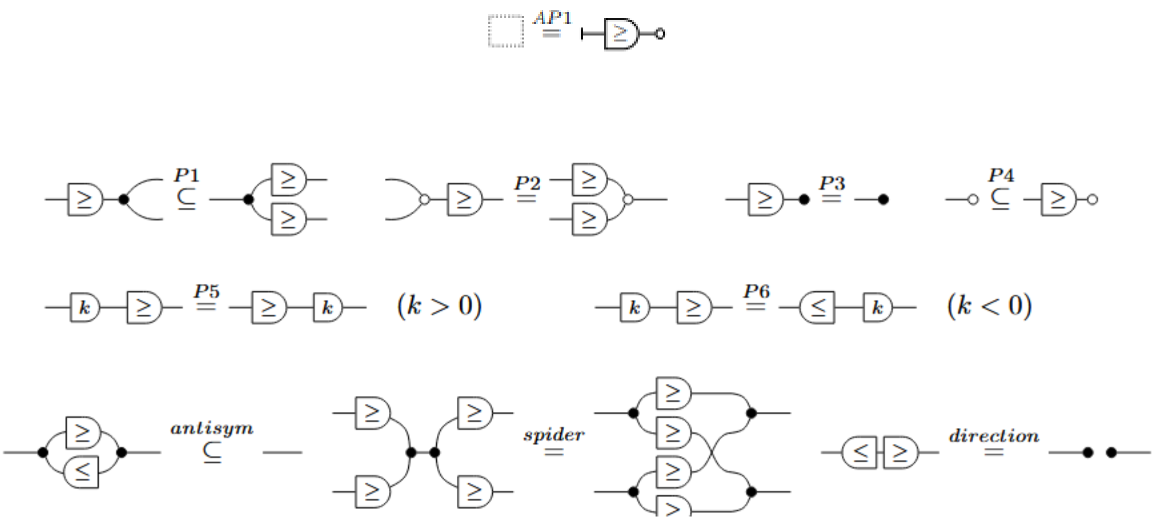
3.2 SYNTAX: THE ALGEBRA $\mathbf{GPA}_{\vdash}$

Definition 3.3 (Graphical Polyhedral Algebra)**.** *We call* **GPA** *the SMT generated by extending* **GAA** *with the following generator and its semantics:*



$$= \{(x,y) | x, y \in k, x \geq y\}$$

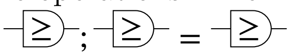
along with the following additional equations:

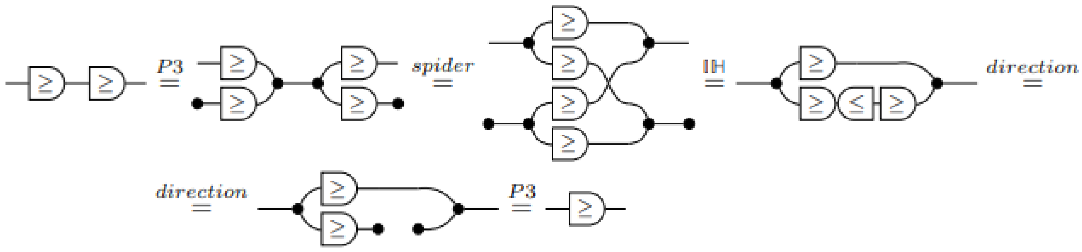


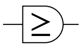
Definition 3.4 (Graphical Polyhedral Cones Algebra)**.** *We call* $\mathbf{GPA}_{\vdash}$ *the SMT generated by* $\mathbf{GPA}$ *but excluding the affine generator* $\vdash$ *and its associated equations. This fragment axiomatises polyhedral cones, where no translation is present.*

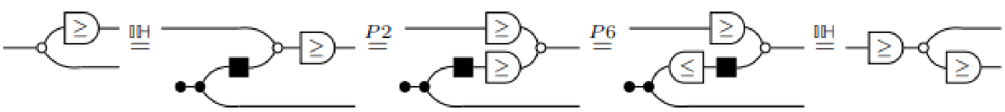
Definition 3.5 (The Free Categories of Polyhedral Diagrams)**.** *We call* $\mathrm{FreeP}_{\mathbf{GPA}}$ *the PROP freely generated by* $\mathbf{GPA}$*, and* $\mathrm{FreeP}_{\mathbf{GPA}_{\vdash}}$ *the PROP freely generated by* $\mathbf{GPA}_{\vdash}$*.*

The central computational primitive for reasoning inside this algebra is Fourier–Motzkin Elimination, which projects a polyhedral cone onto a lower-dimensional space by eliminating one variable at a time.

To derive it diagrammatically, we need some previous results regarding small diagrammatic operations. The first result we prove is the transitivity property of the generator, i.e. 

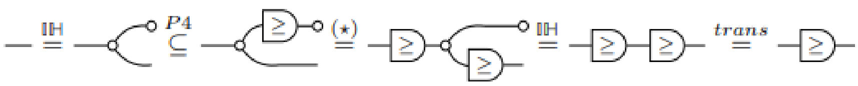


Next, we show that  is reflexive

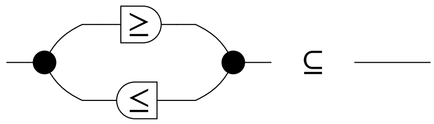


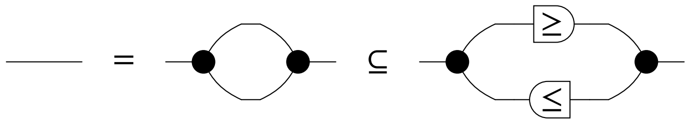
(⋆)

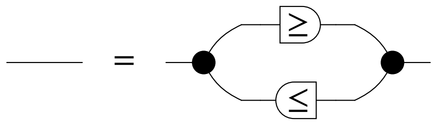
is used below to show that $\geq$ is reflexive:



and antisymetric

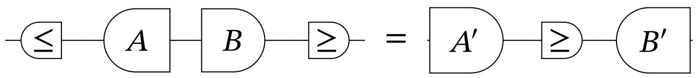
(1) 

(2) 

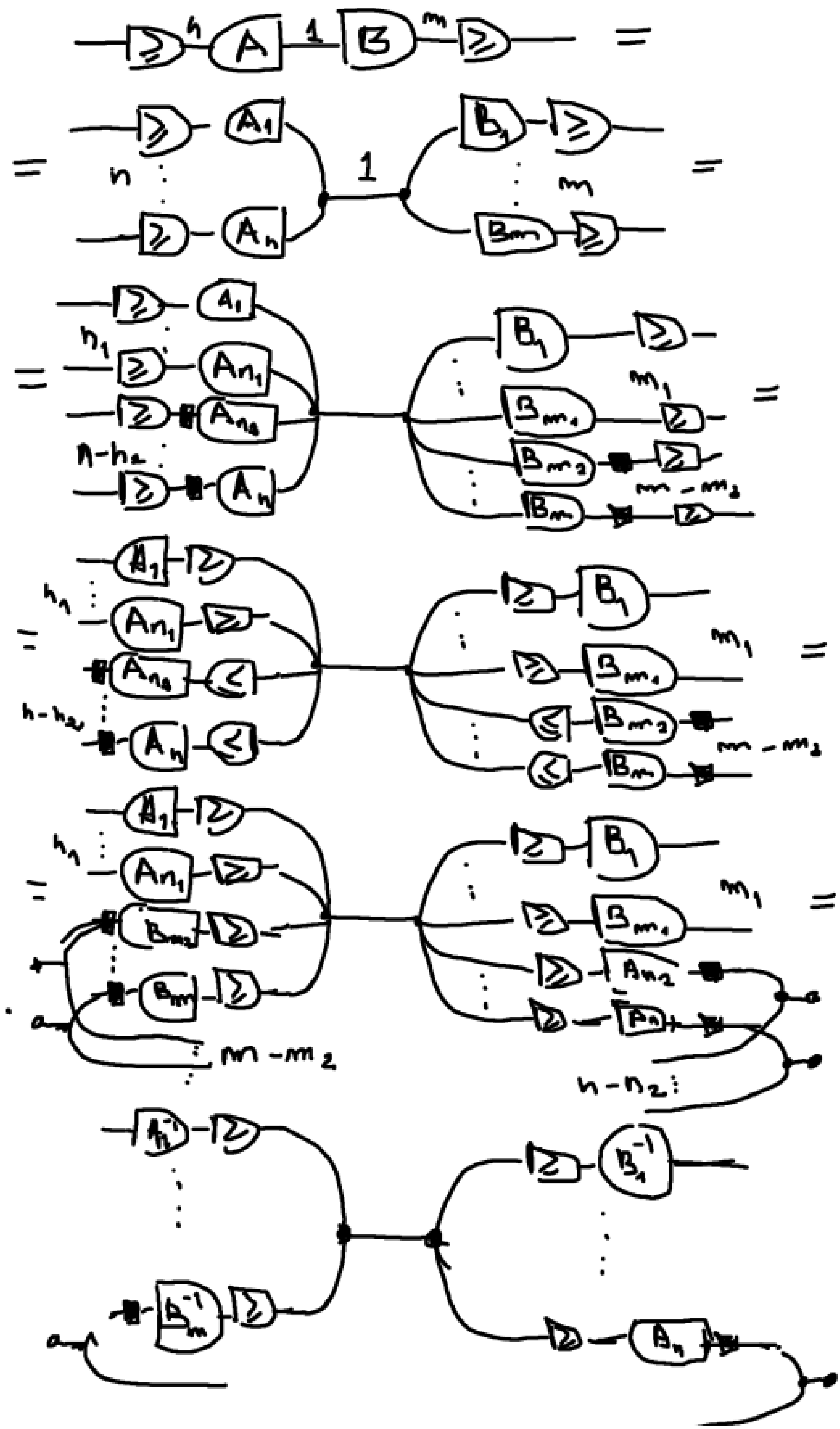
$(1) \wedge (2) \implies$ 

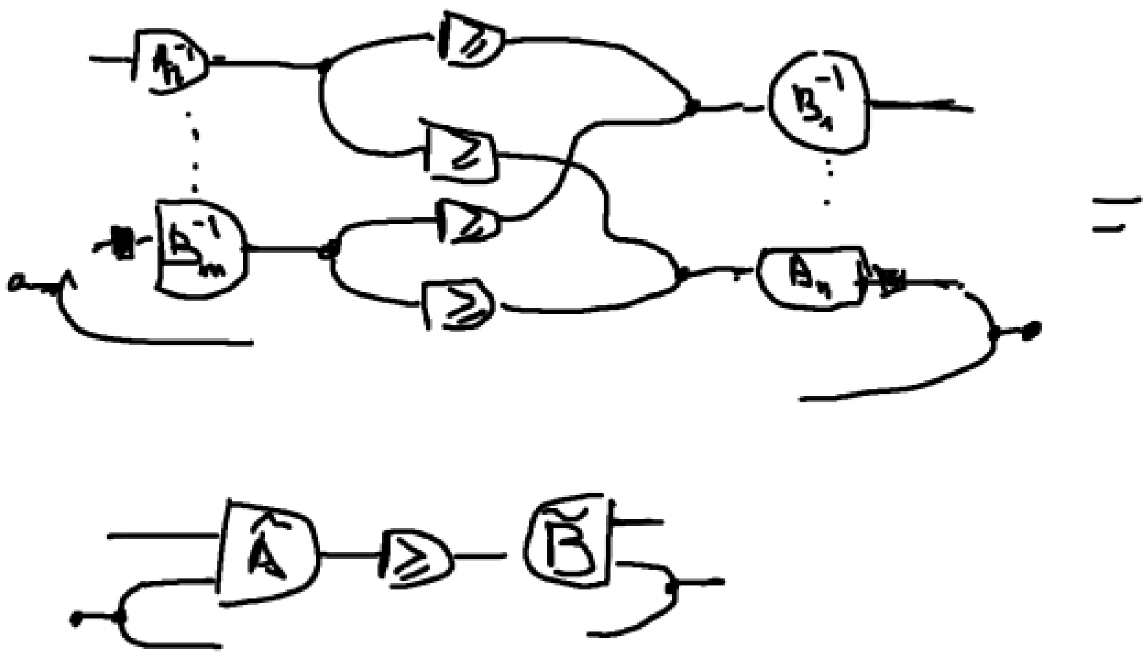
With those results in places we are ready to prove the diagrammatic FME:

Proposition 3.1 (Fourier-Motzkin Elimination)**.** *The following diagrammatic procedure is equivalent to perform the FME algorithm*



Proof. The proof proceeds by induction on the number of inequalities.

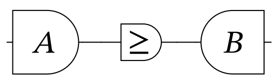




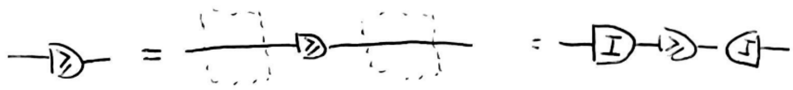
Now we show the inductive step for $n+1$: □

With FME available as a diagrammatic operation, we can now prove the normal form theorem for $\mathbf{GPA}_{\vdash}$. We proceed by structural induction, handling generators, composition, and tensor product in turn, using FME steps to push all inequality generators into a canonical block form.

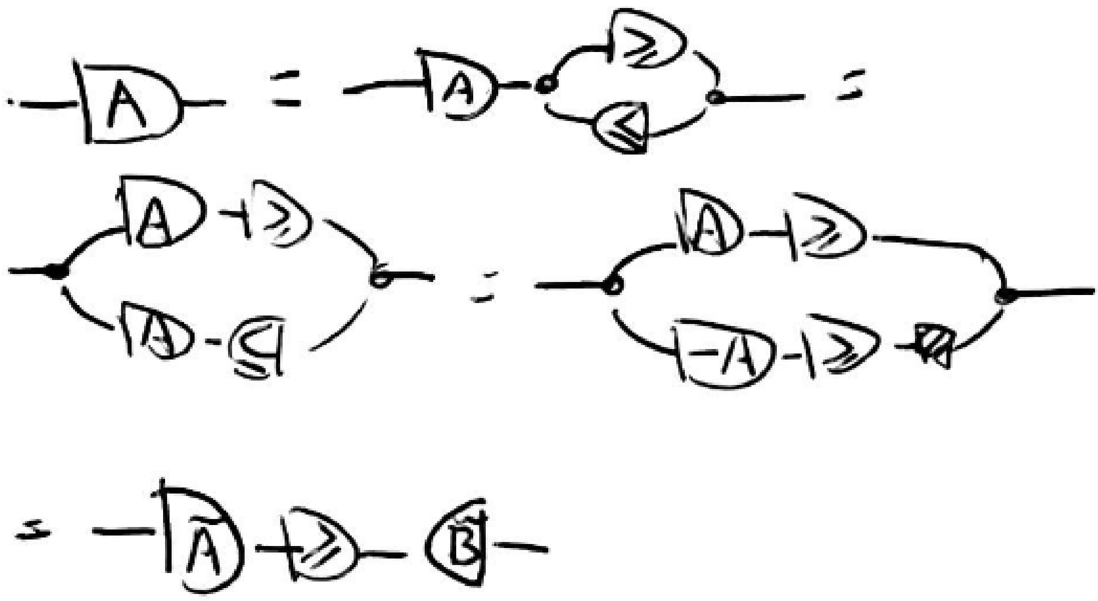
Theorem 3.1 (First Normal Form)**.** *Every diagram in* $\mathbf{GPA}_{\vdash}$ *admits an isomorphic representation, modulo the axioms, of the following form:*



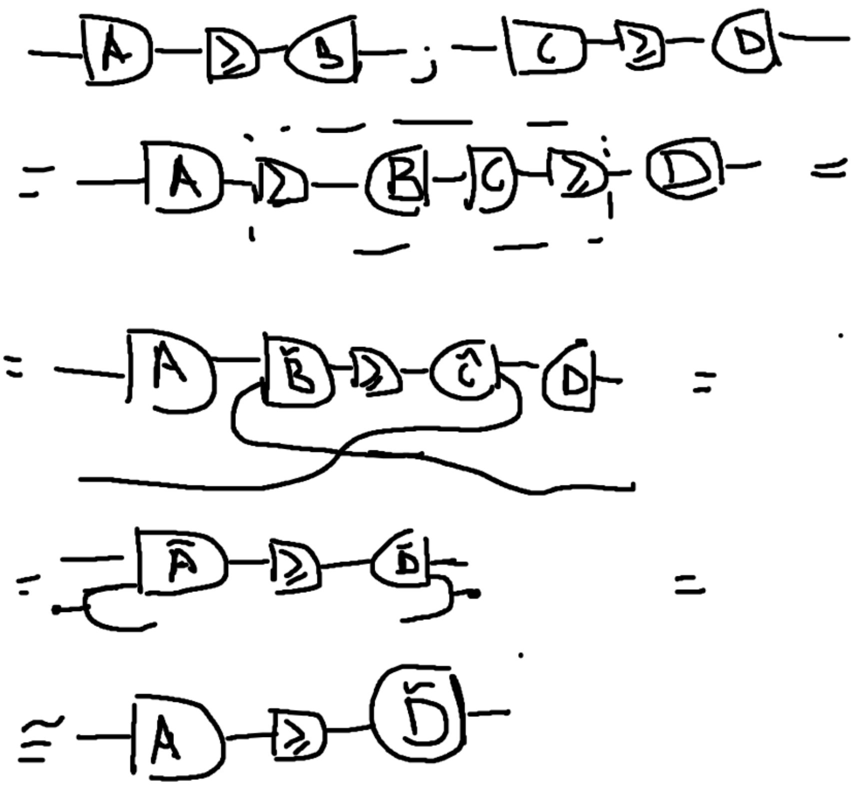
Proof. We proceed by structural induction. The two base cases are as follows. First, the $\geq$ generator:



Second, for any diagram c in **GLA**:



For the inductive step, we consider composition of two diagrams $c;d$ in $\mathbf{GPA}_{\vdash}$. By the inductive hypothesis, both c and d are already in normal form:



And for the tensor product:

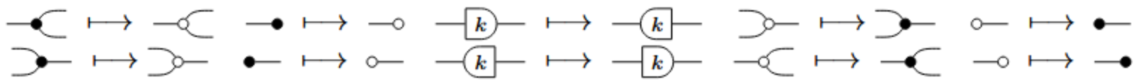
□

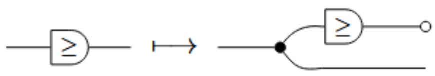
Working with a normal form defined up to isomorphism is in principle weaker than requiring syntactic equality of terms. This choice is motivated by the compact-closed structure, which provides a canonical bijection $\mathrm{Hom}(A, B) \cong \mathrm{Hom}(\mathbb{1}, A^* \otimes B)$, allowing input wires to be treated symmetrically with output wires but only up to this isomorphism, not up to equality. As shown in [BDS21], the same completeness argument can be carried out using strict diagrammatic equality; however, the derivations become significantly more involved. Since every result established for the isomorphic normal form carries over to the strict one, the only difference being the need to bend wires at each step, we adopt the more convenient convention throughout.

This idea was already encountered in the treatment of Graphical Quadratic Algebra [Ste+24], but it is worth pausing to appreciate its significance. The compact-closed structure does not merely allow a more convenient notation; it enables diagrammatic reasoning to serve as a rigorous replacement for coordinate-based algebraic manipulation while remaining extremely intuitive: bending a wire to the left or to the right does exactly what one would expect. Even more, a statement that would classically require several lines of matrix algebra, tracking indices, transposing, and substituting, can instead be verified by a sequence of local wire manipulations, each justified by a single axiom.

This first normal form expresses every diagram as a system of inequalities applied to a linear map. A dual normal form can be obtained by applying the polar operator, which we now define. The polar operator plays the role of duality in the polyhedral setting, exchanging the H-representation (intersection of halfspaces) with the V-representation (conic hull of generators), in direct analogy with the Farkas lemma [BDZ21].

Definition 3.6 (Polar Operator)**.** *The* polar operator $-^\circ$ *is the involution on generators defined by:*

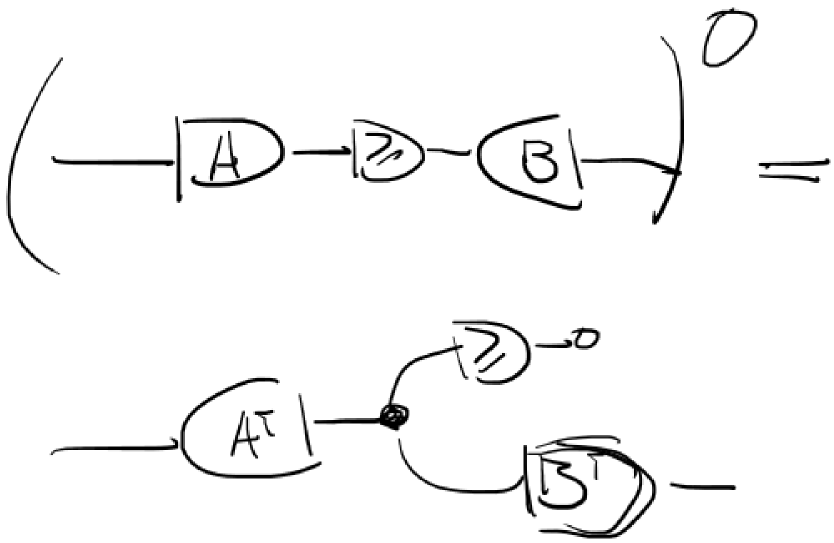




$(c\,;d)^\circ = c^\circ\,;d^\circ \quad (c \oplus d)^\circ = c^\circ \oplus d^\circ$

Its action on diagrams is defined functorially by $(a \circ b)^\circ = a^\circ \circ b^\circ$.

Proposition 3.2 (Second Normal Form)**.** *Every diagram* $s \in \mathbf{GPA}_\vdash$ *can be written as:*



Proof. Apply the polar operator to the first normal form of every diagram $c \in \mathbf{GPA}_\vdash$. Since the polar operator is an involution and acts functorially, the result is again a valid diagram in $\mathbf{GPA}_\vdash$ in the claimed form. □

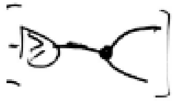
The second normal form expresses every diagram as a conic combination of column vectors, which is precisely the V-representation of a polyhedral cone. This is the representation that will drive the completeness proof: two diagrams that are semantically equal must have the same conic hull, and the axioms of $\mathbf{GPA}_\vdash$ are exactly sufficient to witness this equality diagrammatically.

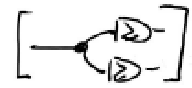
3.3 PRESENTATION RESULTS FOR **PCREL**

We now develop the presentation theory for the homogeneous case, proceeding to establish the three properties that together yield the presentation of **PCRel**.

Lemma 3.1 (Soundness)**.** *If* $s = t$ *in* $\mathrm{FreeP}_{\mathbf{GPA}_\vdash}$*, then* $[s] = [t]$ *in* **PCRel**.

Proof. All axioms except those involving the newly introduced $\geq$ generator are already present in **GAA** and therefore verified. It suffices to check that the new axioms preserve the semantics:

 $= \{(x, \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}) \mid x \geq y_1 \wedge x \geq y_2 \wedge y_1 = y_2\}$

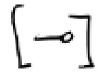
$\subseteq \{x, \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} \mid x \geq y_1 \wedge x \geq y_2\} =$ 

 $= \{\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}, y \mid x_1 + x_2 \geq y\} =$

$\{\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}, y \mid \exists c,d \; x_1 + x_2 \geq c + d \wedge c + d = y\} =$

$\{\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}, y \mid \exists \hat{c}, \hat{d} \; x_1 \geq \hat{c} \wedge x_2 \geq \hat{d} \wedge \hat{c} + \hat{d} = y\} =$ 

 $= \{(x, \bullet) \mid \exists c \; x \geq c\} \overset{\text{Tautology}}{=} \{(x, \bullet) \mid x \in \mathbb{R}\} =$ 

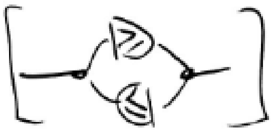
 $= \{(x, \bullet) \mid x = 0\} = \{(x, \bullet) \mid x \geq 0 \wedge x = 0\}$

$\subseteq \{(x, \bullet) \mid x \geq 0\} =$ 

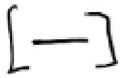
$>0)$  $= \{(x,y) \mid kx \geq y \wedge k > 0\} = \{(x,y) \mid x \geq \frac{y}{k} \wedge k > 0\}$

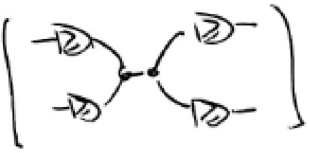
$= \{(x,y) \mid \exists c \; x \geq c \wedge kc = y \wedge k > 0\} =$ 

$k < 0)$  = Same a above, flip the $\geq$ = 

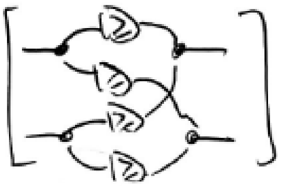
 $= \{(x,y) \mid x \geq y \wedge x \leq y\} =$

$\{(x,y) \mid x = y \wedge x,y \in \overline{\mathbb{R}}\} \subseteq \{(x,y) \mid x = y \wedge x,y \in \overline{\mathbb{R}}\}$

$=$ 

 $= \left\{\left(\begin{pmatrix}x_1\\x_2\end{pmatrix},\begin{pmatrix}y_1\\y_2\end{pmatrix}\right) \mid \exists z\ x_1 \geq z \wedge x_2 \geq z \wedge z \geq y_1 \wedge z \geq y_2\right\}$

$= \left\{\left(\begin{pmatrix}x_1\\x_2\end{pmatrix}\begin{pmatrix}y_1\\y_2\end{pmatrix}\right) \mid x_1 \geq y_1 \wedge x_1 \geq y_2 \wedge x_2 \geq y_1 \wedge x_2 \geq y_2\right\}$

$=$ 

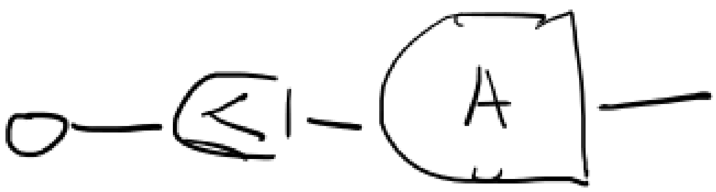
 $= \{(x,y) \mid \exists z\ z \geq y \wedge z \geq x\} =$

$= \{(x,y) \mid x,y \in \overline{\mathbb{R}}\} =$ 

□

Lemma 3.2 (Expressivity)**.** *For every $f : m \to n$ in* **PCRel** *there exists a Σ-term $s : m \to n$ in* $\mathrm{Hom}_{\mathrm{FreeP}_{\mathbf{GPA}_\vdash}}(m,n)$ *such that* $[s] = f$.

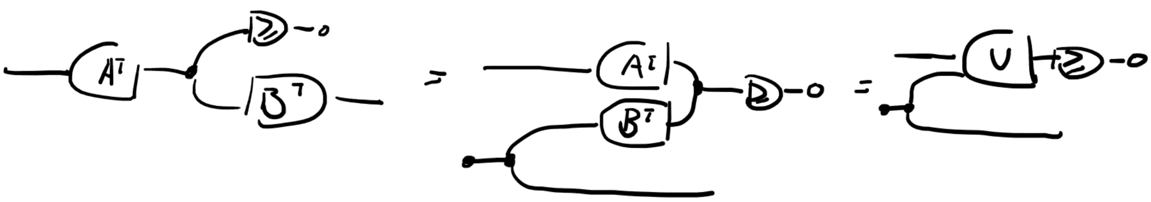
Proof. By Definition 3.1, every polyhedral cone has the form $\{x \mid Ax \geq 0,\ x \in \mathbb{R}^n\}$. This is precisely the interpretation of the diagram:



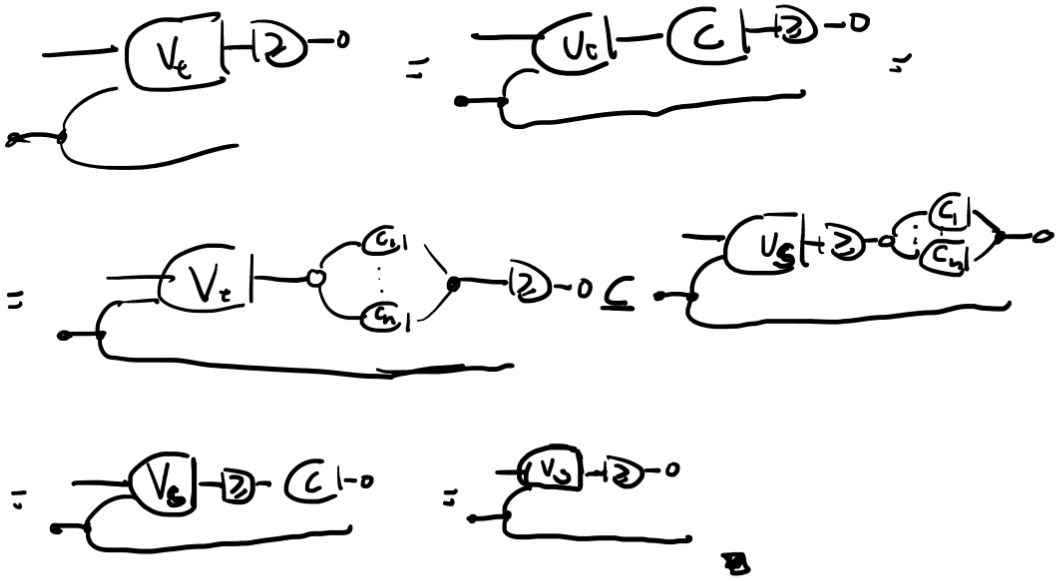
□

Lemma 3.3 (Completeness)**.** *If* $[s] = [t]$ *in* **PCRel***, then* $s = t$ *in* $\mathrm{FreeP}_{\mathbf{GPA}_{\vdash}}$.

Proof. By the second normal form, we write s and t in V-representation form:



The semantics of such a diagram is the conic hull of the column vectors of V. Since $[s] = [t]$ implies in particular $[s] \subseteq [t]$, the conic combinations of V_t are a subset of those of V_s. Hence there exists a non-negative matrix C such that $CV_t = V_s$. The diagrammatic witness is:



The inclusion $[s] \subseteq [t]$ follows by the same argument, yielding equality. □

Theorem 3.2. **PCRel** *is presented by* $\mathrm{FreeP}_{\mathbf{GPA}_{\vdash}}$.

Proof. Soundness, expressivity, and completeness were established by the preceding three lemmas. Together they imply that the canonical PROP morphism is an isomorphism. □

3.4 SEMANTICS: THE PROP **POLYREL** OF POLYHEDRA

Now we start the presentation of the broader class of polyhedral relations by recalling the definition of a polyhedron.

Definition 3.7. *A set $P \subseteq \mathbb{R}^n$ is called a* polyhedron *if there exist $A \in \mathbb{R}^{m\times n}$ and $b \in \mathbb{R}^m$ such that*

$$P = \{x \in \mathbb{R}^n \mid Ax \geq b\}.$$

Note that a polyhedral cone is a special case of a polyhedron with $b = 0$. The distinction is algebraically significant: cones are closed under non-negative scaling and admit a purely homogeneous representation, while general polyhedra may be translated. This distinction is reflected in the two graphical algebras we now define, where $\mathbf{GPA}_{\vdash}$ handles the homogeneous case and **GPA** extends it with an affine generator.

In that light, one can define the category of polyhedral relations as:

Definition 3.8 (The prop **PolyRel** of polyhedra)**.** *Let k be an ordered field. The prop* $\mathbf{PolyRel}_k$ *is defined as follows:*

- *Objects are natural numbers $n \in \mathbb{N}$, interpreted as the space k^n.*
- *A morphism $n \to m$ is a polyhedron $R \subseteq k^n \times k^m$ for which there exist $p \in \mathbb{N}$, $A \in k^{p\times(n+m)}$, and $b \in k^p$ such that*

$$R = \left\{ (x,y) \in k^n \times k^m \;\middle|\; A\begin{pmatrix} x \\ \\ y \end{pmatrix} + b \geq 0 \right\}.$$

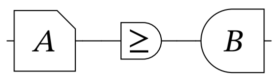
- *Composition is relational composition.*
- *The monoidal product on objects is addition $n \otimes m := n + m$, and on morphisms is the Cartesian product.*

Notice how, the same way that **PCRel** offered us a more relational way to think about Polyhedral Cones, so does **PolyRel**. Indeed, one may think of polyhedrals as closed spaces in some geometry that respects some linear constraints. We argue, however, that this definition is overly convoluted, and a more natural way to denote such objects is to think in terms of relations R, such that $xRy \iff Ax + b \geq By + c$ for some matrices A, B and vectors b, c. Once again, we move from a set-theoretic heavy description to a more logical one.

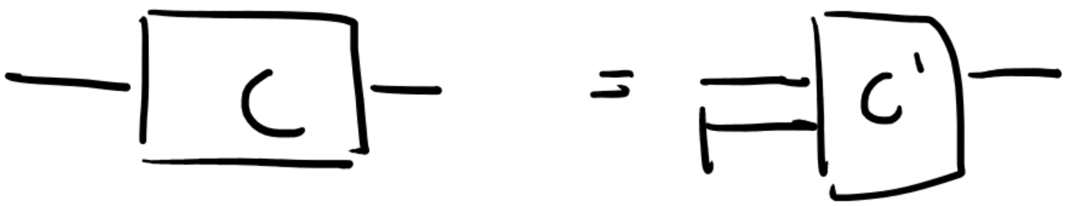
3.5 SYNTAX: THE ALGEBRA **GPA**

We now extend the theory from polyhedral cones to general polyhedra by reintroducing the affine generator $\vdash$. Geometrically, this corresponds to allowing translations: a polyhedron is a translated polyhedral cone, and the generator $\vdash$ provides the syntactic means to encode the right-hand side vector b in $Ax \geq b$. The normal form, soundness, and expressivity arguments all extend smoothly from the homogeneous case.

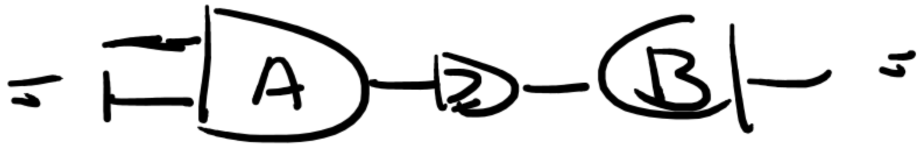
Theorem 3.3 (Normal Form)**.** *For every diagram* c *in* **GPA***, there exists a diagram* d *such that* $\mathsf{c} = \mathsf{d}$ *modulo the axioms and* d *is of the form:*

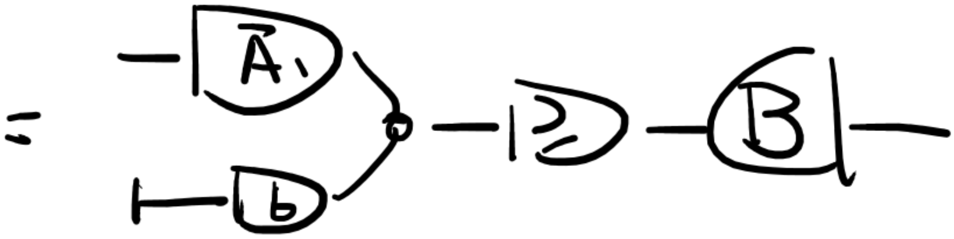


Proof. The derivation proceeds directly. Let c be a diagram of **GPA**. Then we can collect all $\vdash$ on the left such that c' is a diagram of $\mathbf{GPA}_{\vdash}$



From Theorem 3.1 we know that c' admits a normal form in terms of matrices A and B. Breaking down the matrix A in columns we get:





thus the thesis. □

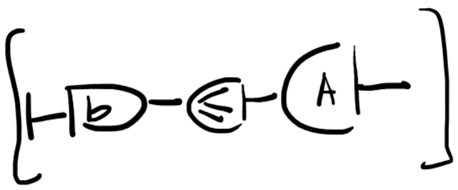
3.6 PRESENTATION RESULTS FOR **POLYREL**

Lemma 3.4 (Soundness)**.** *If* $s = t$ *in* $\mathrm{FreeP}_{\mathbf{GPA}}$*, then* $[s] = [t]$ *in* **PolyRel**.

Proof. This follows immediately from the soundness of **PCRel**. Since the generator $\vdash$ introduces no new axioms beyond those already verified for **PCRel** and **GAA**, and both of those theories are sound in **PolyRel**, soundness carries over directly. □

Lemma 3.5 (Expressivity)**.** *For every* $f : m \to n$ *in* **PolyRel** *there exists a* Σ*-term* $s : m \to n$ *in* $\mathrm{Hom}_{\mathrm{FreeP}_{\mathbf{GPA}}}(m, n)$ *such that* $[s] = f$.

Proof. By Definition 3.7, every polyhedron has the form $\{x \mid Ax \geq b\}$. Appealing to this representation directly, the polyhedron is expressed as the interpretation of the following diagram, where the generator $\vdash$ encodes the right-hand side vector b:

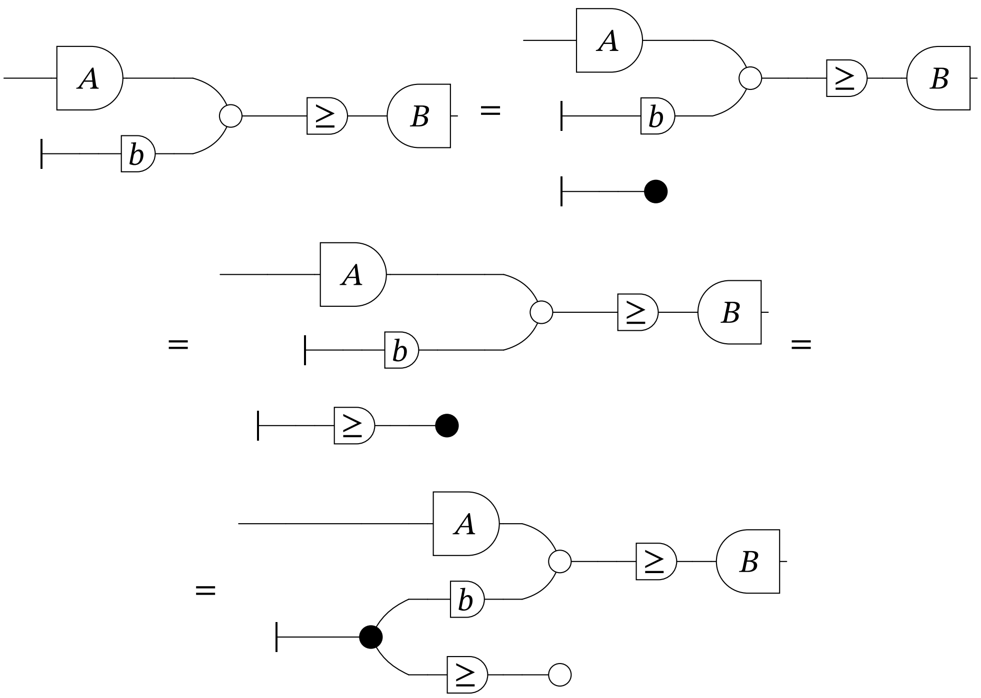
 $= \{(1,x) \mid Ax \geq b,\ x \in \mathbb{R}^n\}$

□

Lemma 3.6 (Completeness)**.** *If* $[s] = [t]$ *in* **PolyRel***, then* $s = t$ *in* $\mathrm{FreeP}_{\mathbf{GPA}}$.

Proof. Every polyhedron P can be homogenised by introducing a slack variable, reducing the affine case to the conic one $P^H = \{(x, y) \in k^{n+1} | Ax + by \geq 0, y \geq 0\}$ which admits a normal form (Theorem 3.1). Under this homogenisation, for each two diagrams $s, t \in \mathbf{GPA}$ with $[s] = [t]$ in **PolyRel**, there are s', t' in $\mathbf{GPA}_{\vdash}$ such that $[s'] = [s]^H, [t'] = [t]^H$. Since two polyhedrals P, Q are equal iff $P^H = Q^H$, then s and t have equal semantics in **PCRel**. Completeness of $\mathbf{GPA}_{\vdash}$ then gives equality of the homogenised diagrams, and de-homogenising recovers equality of s and t in $\mathrm{FreeP}_{\mathbf{GPA}}$.

Diagrammatically this goes as follows:



Whereas the interpretation of the last diagram is $\{(x, y, z) \in k^m | Ax + by \geq Bz, y \geq 0\}$, which is the same as P^H. Therefore, the semantics of s' is that of the homogenisation of $[s]$ and follows the above argument. □

Theorem 3.4. PolyRel *is presented by* $\mathrm{FreeP}_{\mathbf{GPA}}$.

Proof. By the preceding three lemmas. □

Remark 3.1 (Invariance under non-negative matrices). *Whereas the linearity axiom shows the linear generator to be exactly invariant under column-stochastic reweightings (Remark 4.1), the order generator* ge *enjoys a related but weaker property with respect to non-negative matrices. Let $A \in \mathbb{R}^{n \times m}$ with $A_{ij} \geq 0$ for all i, j. If $x, y \in \mathbb{R}^m$ satisfy $x \geq y$ componentwise, then $Ax \geq Ay$ componentwise: for each row i,*

$$(Ax)_i - (Ay)_i = \sum_j A_{ij}(x_j - y_j) \geq 0,$$

since every term is a product of a non-negative entry and a non-negative difference. Diagrammatically, this is the inclusion

$$\mathrm{matrix}[A] \,;\, \mathrm{ge}^{\otimes n} \;\subseteq\; \mathrm{ge}^{\otimes m} \,;\, \mathrm{matrix}[A]^{\otimes 2},$$

verified directly in the semantics of **PolyRel***, exactly as the axioms of* **GPA** *are themselves verified. Unlike Remark 4.1, this is only an inclusion and not an equality: a non-negative matrix need not reflect the order, only preserve it, so* $Ax \geq Ay$ *can hold even when* $x \geq y$ *fails (take* $A = 0$*). In other words, a non-negative matrix always goes through the order generator, certifying that any inequality derived in* **GPA** *remains valid after left multiplication by* A*.*

4 THE LINEAR PROGRAMMING CATEGORY

The theory developed so far has extended the categorical framework by adding a new class of morphisms. We now introduce the final and most expressive theory of the thesis: a graphical algebra for parametric linear programs. Linear programs combine both the inequality structure of polyhedra and the notion of an objective function, and their value functions are precisely the polyhedral convex functions. The central result of this section is the presentation of a complete monoidal theory for right-hand side parametric linear programming.

4.1 SEMANTICS: THE PROP **LPREL**

We begin with several classical results that establish the tight relationship between piecewise affine convex functions, polyhedral functions, and parametric linear programs. These will serve as the semantic foundation for the graphical algebra introduced in the following section, and they justify describing the morphisms of this chapter's semantic category, **LPRel**, interchangeably as parametric linear programs or as polyhedral functions.

Definition 4.1 (Piecewise Affine Convex Function)**.** *A function $f : \mathbb{R}^n \to \mathbb{R}$ is called* piecewise affine convex *if there exist finitely many affine functions $a_i^\top x + b_i$, $i = 1, \dots, k$, such that*

$$f(x) = \max_{i=1,\dots,k} \{a_i^\top x + b_i\}.$$

Definition 4.2 (Polyhedral Function)**.** *A function $f : \mathbb{R}^n \to \mathbb{R} \cup \{+\infty\}$ is* polyhedral *if its epigraph*

$$\mathrm{epi}(f) := \{(x, t) \mid t \geq f(x)\}$$

is a polyhedron.

Definition 4.3 (RHS Parametric Linear Program)**.** *A* right-hand side parametric linear program *has value function*

$$v(b) := \inf_x \{c^\top x \mid Ax \geq b\}.$$

The following two propositions establish that all three notions above are equivalent descriptions of the same class of functions. This equivalence is the semantic cornerstone of the section: it is what allows us to treat polyhedral functions and parametric linear programs as morphisms in the same category.

Proposition 4.1. *A function is piecewise affine convex if and only if it is a polyhedral function.*

Proof. (⇒) Let $f(x) = \max_{i=1,\dots,k}(a_i^\top x + b_i)$. Then

$$\mathrm{epi}(f) = \bigcap_{i=1}^{k} \{(x,t) \mid t \geq a_i^\top x + b_i\},$$

which is an intersection of finitely many halfspaces, hence a polyhedron.

(⇐) If f is polyhedral, its epigraph is a polyhedron, so there exist finitely many affine inequalities $t \geq a_i^\top x + b_i$ such that $f(x) = \max_{i=1,\dots,k}(a_i^\top x + b_i)$. Thus f is piecewise affine and convex. □

Proposition 4.2. *A function $v : \mathbb{R}^m \to \mathbb{R} \cup \{+\infty\}$ is piecewise affine convex if and only if it is the value function of a right-hand side parametric linear program.*

Proof. (⇒) Let $v(b) = \max_{i=1,\dots,k}(y^i)^\top b$ and define $Y := \mathrm{conv}\{y^1, \dots, y^k\}$. Then $v(b) = \max_{y \in Y} b^\top y$. Since Y is a polyhedron admitting the representation $Y = \{y \mid A^\top y = c,\ y \geq 0\}$, linear programming duality gives $v(b) = \inf\{c^\top x \mid Ax \geq b\}$.

(⇐) Let $v(b) = \inf\{c^\top x \mid Ax \geq b\}$. Dualising, $v(b) = \sup\{b^\top y \mid A^\top y = c,\ y \geq 0\}$. Since the feasible set $Y = \{y \mid A^\top y = c,\ y \geq 0\}$ is a polyhedron, the supremum is attained at its extreme points $y^1, \dots, y^k$, yielding $v(b) = \max_{i=1,\dots,k}(y^i)^\top b$. Thus v is piecewise affine convex. □

The following proposition is the key technical lemma for the completeness proof of **GLP**. It states that two parametric linear programs with the same value function must share the same epigraph polyhedron, providing the bridge between semantic equality and polyhedral equality that the completeness argument requires.

Proposition 4.3. *Let*

$$v_i(y) := \inf\{\mathbf{1}^\top x \mid A_i x \geq B_i y\}, \qquad i = 1, 2,$$

where $A_i \in \mathbb{R}^{m_i \times n_i}$ and $B_i \in \mathbb{R}^{m_i \times k}$. If $v_1(y) = v_2(y)$ for all $y \in \mathbb{R}^k$, then both programs admit a common representation through the same epigraph polyhedron:

$$v_i(y) = \inf\{t \mid (y,t) \in E\}, \qquad E := \mathrm{epi}(v_1) = \mathrm{epi}(v_2).$$

Proof. For each $i = 1, 2$, the epigraph of v_i satisfies

$$(y,t) \in \mathrm{epi}(v_i) \iff \exists x : A_i x \geq B_i y,\ \mathbf{1}^\top x \leq t,$$

so $\mathrm{epi}(v_i)$ is the projection onto (y,t) of the polyhedron $P_i = \{(x,y,t) \mid A_i x \geq B_i y,\ \mathbf{1}^\top x \leq t\}$. Projections of polyhedra are polyhedra, so each $\mathrm{epi}(v_i)$ is a polyhedron. Since $v_1 = v_2$ pointwise, we have $\mathrm{epi}(v_1) = \mathrm{epi}(v_2) =: E$. Setting $E = \{(y,t) \mid Cy + dt \geq 0\}$, we obtain $v_i(y) = \inf\{t \mid Cy + dt \geq 0\}$ for both $i = 1, 2$. □

With the equivalences above in place, we now define the semantic category of this chapter. **LPRel** describes morphisms as parametric linear programs; by the propositions above, these coincide with polyhedral functions, and we will use both descriptions interchangeably according to which is more convenient.

Definition 4.4 (The prop **LPRel** of Linear Programs)**.** *The prop* **LPRel** *is defined as follows:*

- *Objects are natural numbers* $n \in \mathbb{N}$*, interpreted as the space* $\mathbb{R}^n$*.*
- *A morphism* $n \to m$ *is a right-hand side parametric linear program with value function*

$$(x, y) \mapsto \inf_z \{c^\top z \mid Az \geq B[x\ y]^\top\},$$

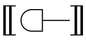
  where $x \in \mathbb{R}^n$ *and* $y \in \mathbb{R}^m$*.*

- *Composition is given by minimisation over the shared variable.*
- *The monoidal product on objects is addition* $n \otimes m := n + m$*, and on morphisms is direct sum.*

4.2 SYNTAX: THE ALGEBRA **GLP**

With the semantic equivalences established, we now introduce the final graphical algebra of the thesis. The algebra **GLP** extends **GPA** with a single new generator whose semantics encodes the linear objective function $c^\top x$. Just as the gray arrow in **GQA** lifted the theory from binary relations to weighted relations by "measuring" the ℓ_2 norm, this new generator lifts **GPA** from polyhedral constraints to full linear programs by attaching a linear cost to the feasible region.

Definition 4.5 (**GLP**)**.** *We call* **GLP** *the symmetric monoidal theory obtained by extending* **GPA** *with the following generator:*

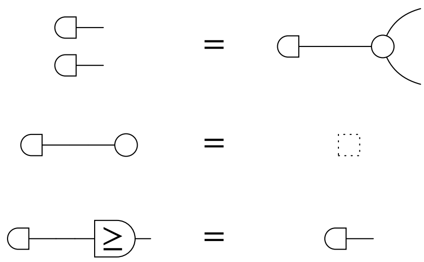
$[\![$  $]\!] = (\bullet, x) \mapsto x$

together with the following axioms governing its interaction with the existing generators:

(linearity)

(zeroeq)

(inf)



The axioms represent expected properties of the linear cost function. The one on the left express the linear property $f(x) + f(y) = f(x + y)$. Observe that f is the interpretation of the ᗡ − and the sum denotes the vertical composition; composing two ᗡ − vertically (adding) is the same as adding the wires in a single ᗡ −. This axiom is also the fundamental property of LPs, which is the fact that they are invariant to column stochastic matrices (see 4.1). In turn, the equation on the right express the fundamental property of minimization, the infimum of any quantity lies in its lower bound.

$$\inf_x \{x | x \geq y\} = y$$

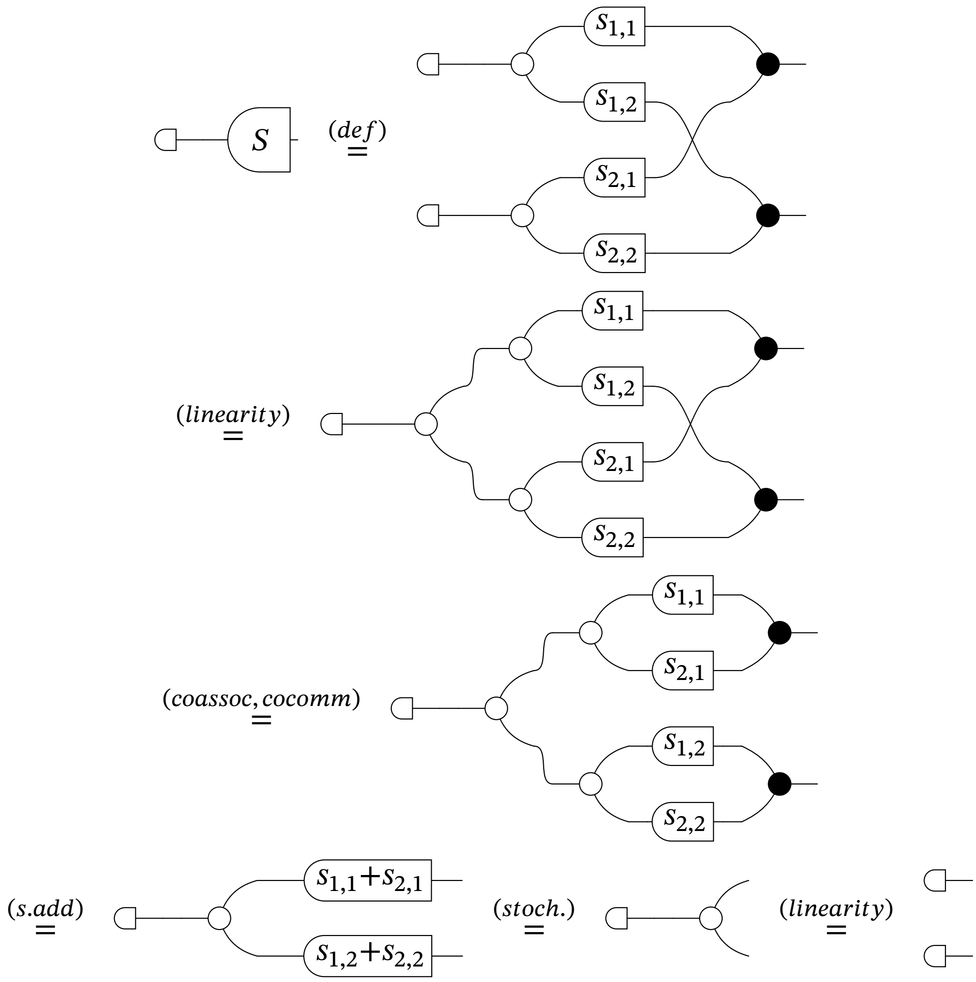
Finally, the axiom at the middle is the tautological case.

$$inf 0 = 0$$

Remark 4.1 (Invariance under column-stochastic matrices)**.** *The linearity axiom encodes the fundamental invariance of linear programs under column-stochastic reweightings of the objective: for any* $n \times m$ *matrix* S *whose columns each sum to one,*

$$ᗡ\!\!-^{\otimes n};\ S^{op} = ᗡ\!\!-^{\otimes m},$$

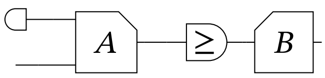
where S^{op} *denotes the comatrix of* S*. Note that only the column sums are used; non-negativity of the entries plays no role. The proof is diagrammatic, we display the proof for a* 2×2 *matrix* S *with* $s_{1,1} + s_{2,1} = s_{1,2} + s_{2,2} = 1$*; every step is an instance of an axiom scheme that applies wire-by-wire, so the same chain of rewrites goes through verbatim for arbitrary* $n \times m$*.*



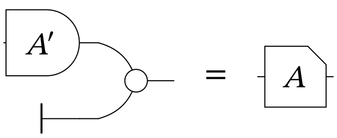
Definition 4.6. *We call* $\mathrm{FreeP}_{\mathbf{GLP}}$ *the PROP freely generated by* **GLP**.

The normal form theorem for **GLP** extends the one for **GPA**: every diagram can be reduced to a **GPA** block encoding the constraints, composed with the new generator encoding the objective. The proof proceeds by structural induction, using the existing normal form for **GPA** as the inductive base.

Theorem 4.1 (Normal Form)**.** *For every diagram* $\mathsf{c} \in \mathbf{GLP}$*, there exists a diagram* c' *in normal form such that* $\mathsf{c} = \mathsf{c}'$*:*

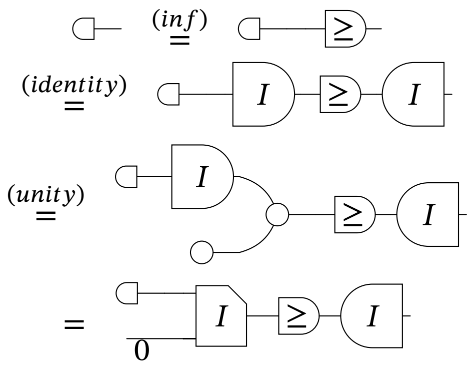


With A *and* b *being an affine transformation. Namely:*

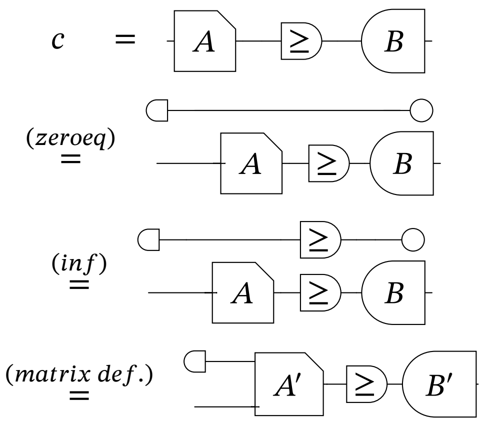


Proof. The proof is by structural induction. First we show the two base cases:

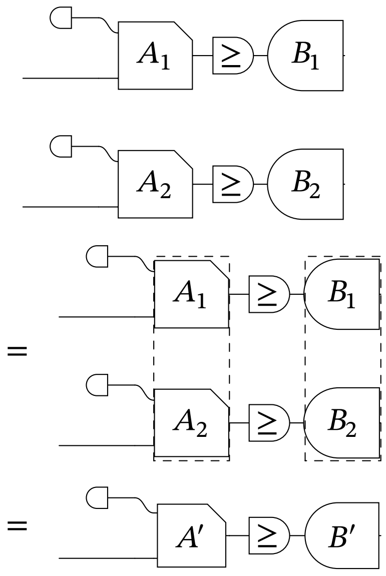
- The generator $\sqsupset -$



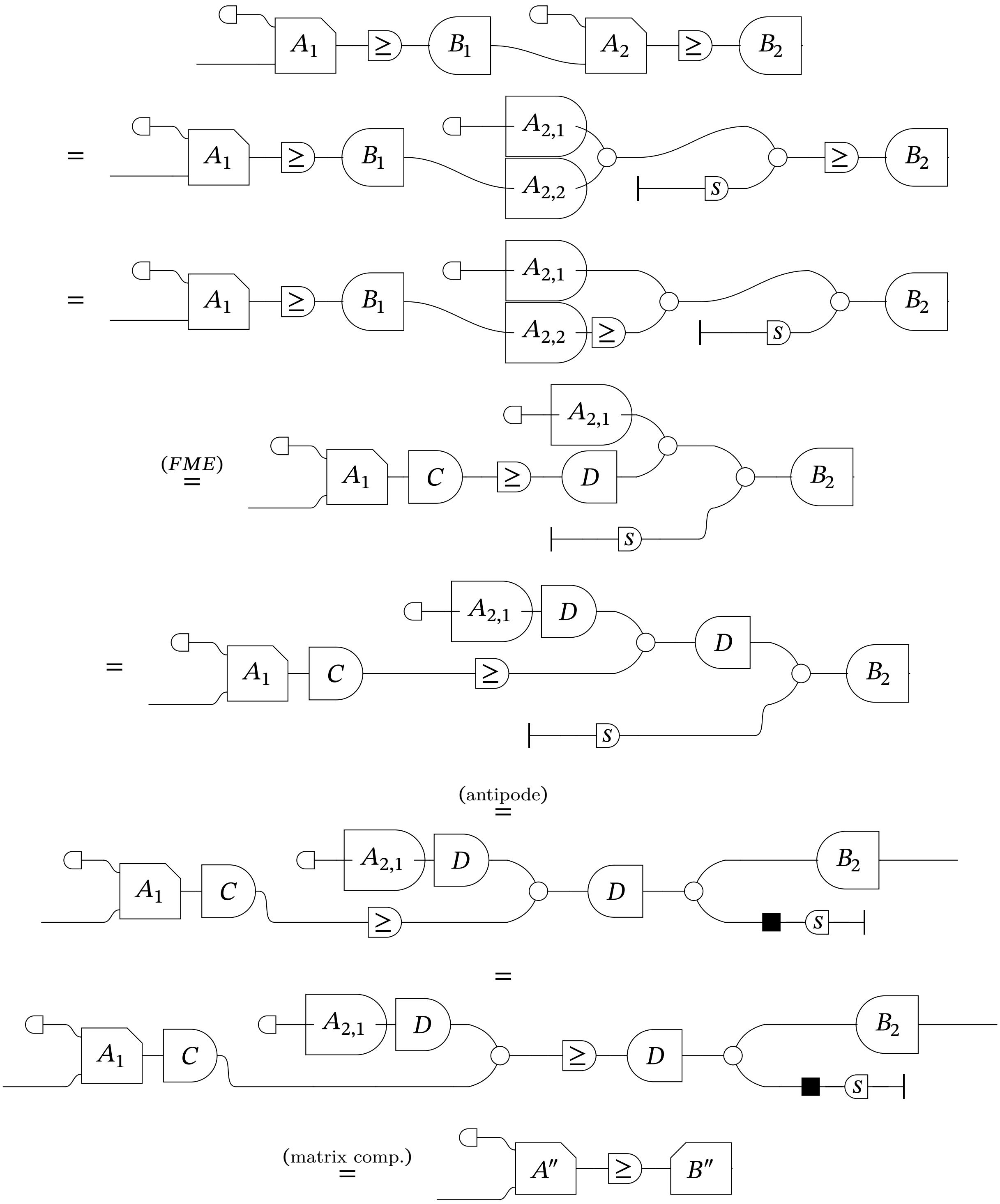
- A diagram $c \in$ **GPA**



For the inductive step, we verify that for any two diagrams $c, d \in$ **GLP** satisfying the inductive hypothesis, both horizontal and vertical composition preserve the normal form. Closure under vertical composition follows by stacking the two normal forms and collapsing the two independent A_i, ge, B_i blocks into a single block-diagonal matrix pair, as shown below:



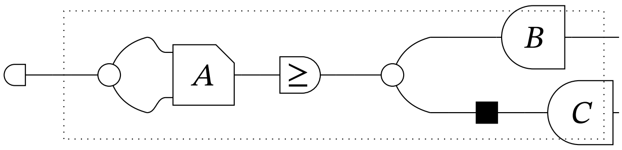
For horizontal composition, the two normal forms are joined along the wire connecting B_1 to A_2; applying Fourier-Motzkin Elimination to this pair recovers a fresh matrix-ge-comatrix triple, after which the two resulting runs of matrices, one in each direction, compose into a single matrix and a single comatrix, recovering the normal form:



□

The following corollary reveals the geometric meaning of the normal form: the **GPA** part of every **GLP** diagram encodes the epigraph of the corresponding value function. This is the key structural insight connecting the syntax to the semantic equivalence of Proposition 4.3.

Corollary 4.1 (Epigraph Form)**.** *Given a diagram* $d \in \mathbf{GLP}$ *in normal form, it can always be rewritten as a diagram* d' *of the form:*



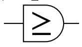
where the semantics of the dotted diagram is

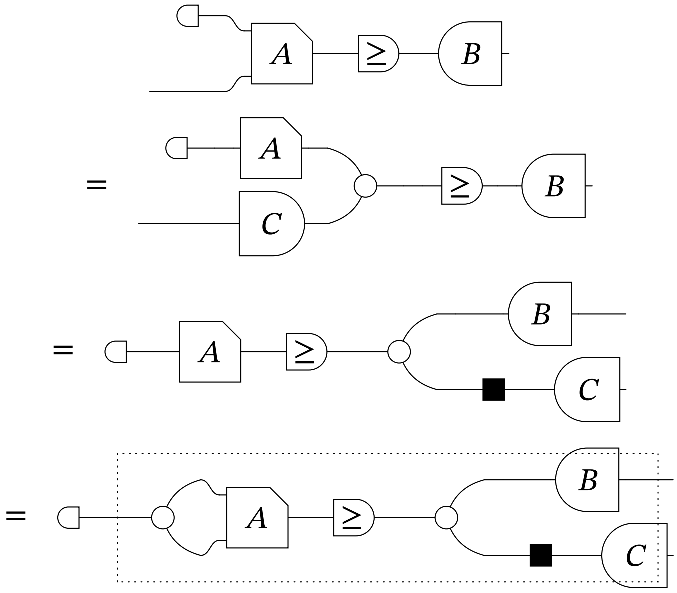
$$\left\{(t, y) \mid \exists\, x,\ A'x + a \geq [B\ \ -C]\, y,\ t \geq x,\ y = \begin{bmatrix} y_1 \\ \\ y_2 \end{bmatrix}\right\}$$

. This is the epigraph of the original diagram d

$$[d](y_1, y_2) = \inf_z\ z \quad \textit{s.t. } Az + a \geq By_1 - Cy_2$$
$$\mathrm{epi}([d]) = \{(t, y_1, y_2) \mid \exists\, z \leq t,\ Az + a \geq By_1 - Cy_2\}$$

such that $A = [A'\ C]$*. We call this the* epigraph form *of a linear program.*

Proof. Starting from d in normal form, split the relation generator along $A = [A'\ C]$ and pull the resulting C-block through the  generator using the antipode axiom:



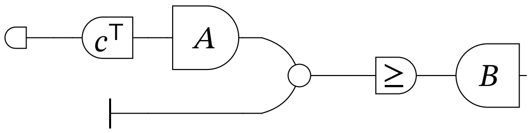
The resulting diagram is precisely the claimed form d', with the dotted box exhibiting epi($[d]$) as a **GPA** diagram. □

4.3 PRESENTATION RESULTS FOR **LPREL**

With the syntax and its normal form in place, we are now ready to prove the three presentation properties for **GLP**.

Lemma 4.1 (Expressivity). *For every $f : m \to n$ in* **LPRel** *there exists a Σ-term $s : m \to n$ in* $\mathrm{Hom}_{\mathrm{FreeP}_{\mathbf{GLP}}}(m, n)$ *such that* $[s] = f$.

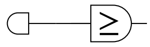
Proof. Since every polyhedral function is the maximum of a finite family of affine functions, it suffices to show that every piecewise affine convex function is expressible in **GLP**. For an arbitrary such function $f(y) = \max_i R_i(y)$, where $\{R_i\}_{i=1,\dots,n}$ is a family of affine functions, the following diagram provides the required syntactic representative:

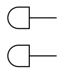
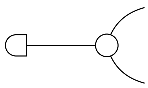
 $[\![=]\!]\ y \mapsto \inf_x c^\top x$ s.t. $Ax + b \geq By$

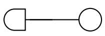
□

Lemma 4.2 (Soundness). *If $s = t$ in* $\mathrm{FreeP}_{\mathbf{GLP}}$, *then* $[s] = [t]$ *in* **LPRel**.

Proof. All axioms inherited from **GPA** are already sound by the results of the previous section. It suffices to verify that the new axioms involving the LP generator preserve the semantics, which follows by direct computation:

(linear.ge)  $[\![=]\!] (\bullet, y) \mapsto y = (\bullet, y) \mapsto \inf_{x \geq y} x [\![=]\!]$ 

(*lin.sum*)  $[\![=]\!]\ (\bullet, \binom{y_1}{y_2}) \mapsto y_1 + y_2 [\![=]\!]$ 

(*lin.zero*)  $[\![=]\!] (\bullet, \bullet) \mapsto \inf_{x=0} x = 0\ [\![=]\!]$ 

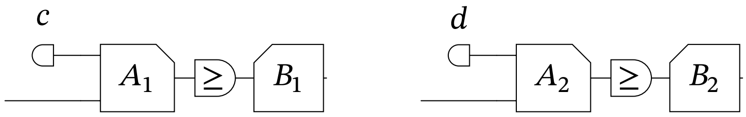
□

The final result of the thesis is the completeness theorem for **GLP**. The proof brings together all the main tools developed throughout: the epigraph form of Corollary 4.1, the shared epigraph property of Proposition 4.3, the completeness of **GPA**, and the FME-based normal form.

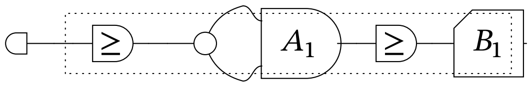
Lemma 4.3 (Completeness). *If $[c] = [d]$ in* **LPRel**, *then $c = d$ in* $\mathrm{FreeP}_{\mathbf{GLP}}$.

Proof. Let $v_1 = [c]$ and $v_2 = [d]$. By assumption $v_1 = v_2$, so Proposition 4.3 gives $\mathrm{epi}(v_1) = \mathrm{epi}(v_2) =: E$.

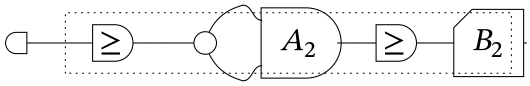
By Corollary 4.1, both c and d admit epigraph forms $c = \sqsubset -; c^*$ and $d = \sqsubset -; d^*$ with $[c^*] = [d^*] = E$ in **PolyRel**. Since c^* and d^* are **GPA** diagrams with the same denotation, completeness of **GPA** gives $c^* = d^*$ in $\mathrm{FreeP}_{\mathbf{GPA}}$; in particular both reduce to the same **GPA** normal form. Composing this common diagram with the leftover $\sqsubset -$ collapses c and d to the same **GLP** diagram:



(epigraph form, Cor. 4.1)

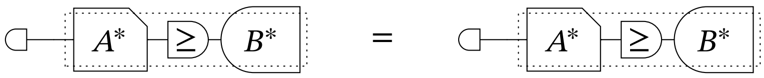


c^*

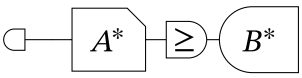


d^*

(the dotted boxes c^*, d^* are **GPA** diagrams with $[c^*] = [d^*] = E$; by **GPA** completeness both reduce to the same normal form)



(composing the common **GPA** normal form with the leftover $\sqsubset -$ yields the same **GLP** diagram)



Therefore $c = \sqsubset -; c^* = \sqsubset -; d^* = d$. □

As a consequence of the three lemmas above:

Theorem 4.2. LPRel *is presented by* $\mathrm{FreeP}_{\mathbf{GLP}}$.

Proof. By the preceding three lemmas. □

Finally, we've built, through the addition of one generator and three axioms, an elegant and complete diagrammatical theory for Parametric Linear Programming. We argue that, such result, not just unlocks a new way to interact with LP problems in a algebraic and computational way, but also provide a new perspective to optimization through the lens of a compositional framework. Whereas optimization, namely LP problems, are lifted above the settings of functional thinking and achieves a new status as a relational-theoretic field embeded into the Extended-Lawvere Quantale category.

The **GLP** theory, in turn, has a natural extension to an even more expressive theory when enriched with the gray arrow generator from **GQA**. We believe that the class of quadratic programs could be, just as was done in this work, fully axiomatized by such broader SMIT. However, such studies are left for future works, along with other possible applications of current results.

5 CONCLUSION

This thesis began with a philosophical claim, that the nature of a mathematical object is fully determined by its relationships to other objects, and that universal constructions reveal connections invisible to any particular viewpoint. Optimization, by contrast, is often presented as a collection of problem-specific techniques. The central claim of the thesis is that these two perspectives are not merely compatible, but that optimization *emerges* from categorical structure naturally, as an intrinsic property of relational composition and is natural to mathematical thinking.

Figure 5.1 illustrates part of this belief; expressing the relationships among all the categorical structures developed in this thesis.

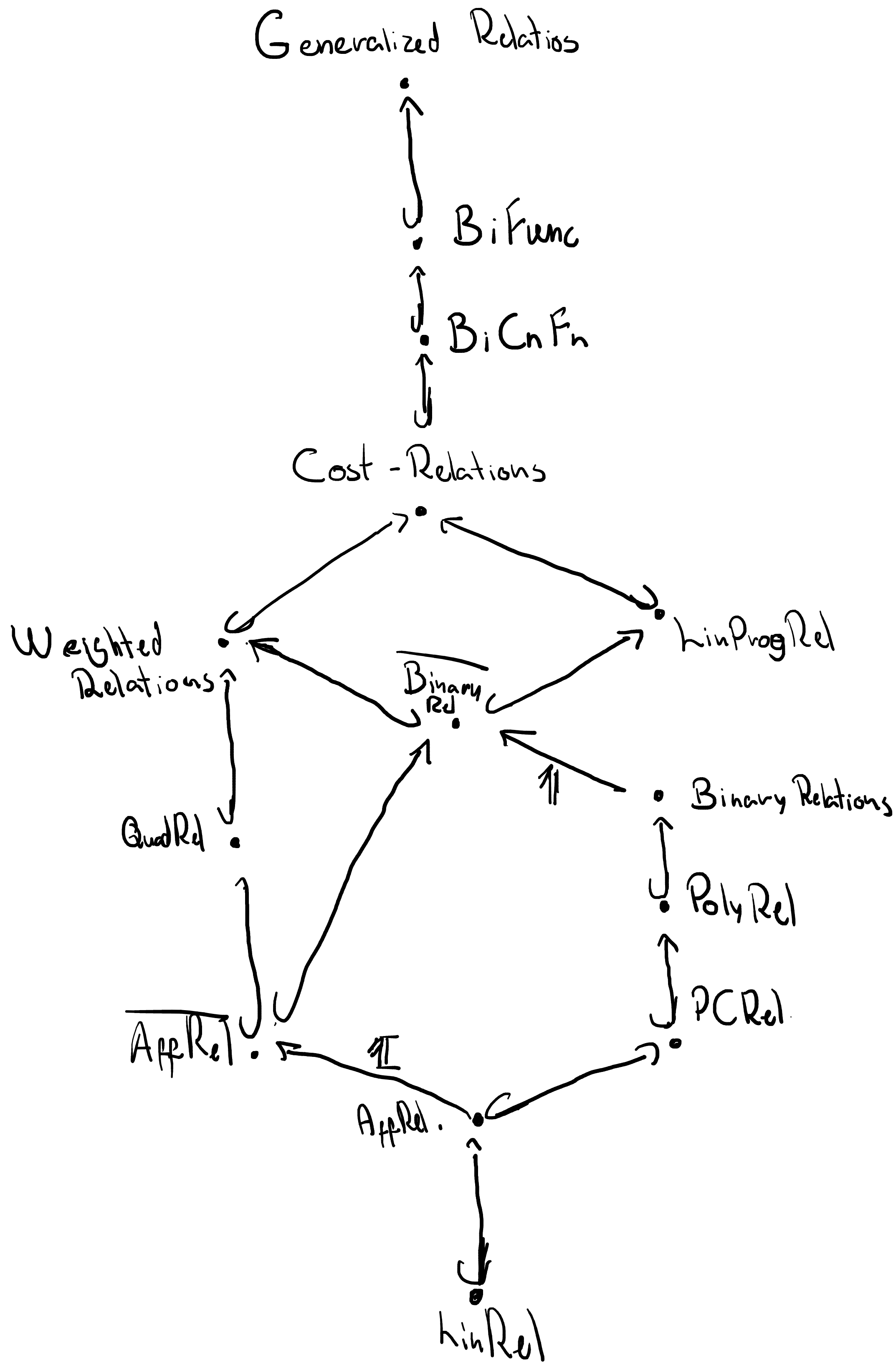


Figure 5.1 – The hierarchy of relational categories developed in this thesis. Each inclusion preserves the monoidal and compositional structure of the layer below.

- Moving upward corresponds to increasing expressive power: each layer generalises the one below by enlarging the class of morphisms, while preserving the compositional structure.
- The morphisms marked $\hookrightarrow$ are inclusion functors, embedding each category as a full subcategory of the one above.
- The morphism marked $\mathbf{1}$ denotes the indicator function embedding, which sends a set-valued relation to its $\{0, +\infty\}$-valued characteristic function.

What this hierarchy makes visible is something that would be difficult to articulate within any single one of its layers. Polyhedral relations and quadratic relations are not merely two classes of optimization problems that happen to admit graphical representations; they are parallel enrichments of the same underlying affine structure, each adding a different kind of expressive power. Their common base in **AffRel** and their common embedding into **Cost-Relations** is not an accident of notation but a theorem, witnessed by the inclusion functors in the diagram. A single compositional framework within which relations, functions, constraints, and objectives are unified as morphisms, and within which optimization is not a technique applied to a problem but an algebraic operation itself.

The broader significance of this perspective lies in what it enables beyond the specific algebras treated here. This opens several directions for future work. Such as further extending our current language to even further and more general problems. A natural next step would be the enrichment of GQA [Ste+24] to reach the state of a SMIT. This should offer an even stronger language to treat Quadratic Problems.

A more applied direction concerns the computational implications of diagrammatic normal forms. If two optimization problems are equal as morphisms in **LPRel** if and only if their diagrams are related by the rewriting rules of **GLP**, then diagram simplification is problem simplification: a rewriting system on string diagrams is simultaneously a symbolic preprocessor for optimization solvers. Such an application of graphical algebras for theorem solvers was already pointed in other works [BDZ21; BDS21].

This thesis has taken one more step on the road to fully expressing optimization within its own algebraic setting. It has shown that the language of string diagrams can absorb, unify, and extend three of the central theories of mathematical optimization within a single categorical framework built incrementally from the ground up. The theories here presented are not three separate pieces wearing categorical clothing, but a single theory, viewed at three different levels of expressive resolution.

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